

AI Exposure and Age-Differentiated Employment: Evidence from Norwegian Register Data

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Abstract

Using Norwegian population administrative data covering 2021 to 2026, we document age-differentiated employment patterns in occupations differentially exposed to artificial intelligence. We construct monthly employment-per-capita series at the cell defined by occupation, age group, month, and sector, indexed to October 2022 (the month before the November 30, 2022 release of ChatGPT). Across four AI exposure measures (theoretical, revealed usage, ability-based, and observed time-weighted), the most-exposed quintile of occupations declines by approximately 10 percentage points for workers aged 22–25, while the least-exposed quintile is roughly flat. In selected high-exposure occupations the decline for 22–25 year-olds reaches 30 percent (software developers, ICT systems analysts). Total employment for 22–25 year-olds is roughly flat over the period, so the most-exposed quintile’s decline reflects reallocation across exposure quintiles rather than net job loss. Workers aged 31 and older show no exposure gradient. The gradient is stronger with theoretical and ability-based exposure than with revealed Claude usage, and stronger when occupations are ranked by Handa automation share than by augmentation share. The pattern is concentrated in the private sector; public-sector employment shows no corresponding gradient. Wages and other intensive-margin outcomes show no divergence by exposure quintile, consistent with adjustment on the hiring margin under Norway’s two-tier bargaining system. The Norwegian patterns match the direction of US, Swedish,

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and UK findings and contrast with the Finnish null; they also complement the Danish finding that workplace AI chatbot adoption does not appear to drive an analogous aggregate decline.

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1 Introduction

Is generative AI reshaping employment patterns? A growing literature investigates a post-2022 age gradient in occupations most exposed to artificial intelligence: employment of young workers declines relative to older workers, while no such gradient appears in less-exposed occupations. [Brynjolfsson et al. \(2025\)](#) find this pattern in US payroll data, [Lodefalk et al. \(2026\)](#) document a similar gradient in Sweden, and [Teeselink \(2025\)](#) reports consistent UK evidence. On the other hand, [Humlum and Vestergaard \(2026\)](#) and [Kauhanen and Rouvinen \(2026\)](#) report null patterns from Denmark and Finland.

We document Norwegian employment patterns. Using population-level register data from the Norwegian employer reporting system for 2021 to 2026, we construct monthly employment-per-capita series at the level of cells defined by occupation, age group, and AI exposure quintile. We normalize employment counts by population in each age group to remove the effect of changing cohort sizes. Across four distinct AI exposure measures, employment of workers aged 22 to 25 in the most-exposed quintile declines relative to the least-exposed quintile after October 2022, the index date used throughout the paper and the month preceding the November 30, 2022 release of ChatGPT. The pattern is concentrated in the private sector. Public-sector employment shows no corresponding age-exposure gradient. Median monthly earnings within occupation-age cells exhibit no divergence by exposure quintile over the same period.

The data cover the entire employed population at monthly frequency, with age assigned at monthly precision. We test the gradient against four exposure measures: a theoretical task measure ([Eloundou et al., 2024](#)), a revealed-usage measure from Claude.ai conversations ([Handa et al., 2025](#)), an ability-based measure ([Felten et al., 2021](#)), and a time-weighted observed-usage measure ([Anthropic, 2026](#)). The age-exposure gradient appears under all four, strongest for the theoretical and ability-based measures and weakest for revealed usage.

Norway is also a setting where AI tools have diffused rapidly. A 2026 Microsoft report ranks Norway third globally in population-level AI usage, after the UAE and Singapore ([Microsoft AI Economy Institute, 2026](#)). OECD data for 2025 place Norway first among 33 European countries for AI use at work ([McQuinn, 2026](#)). At the enterprise level, the share of Norwegian firms (10 or more employees) reporting AI use rose from 11 percent in 2021 to 30 percent in 2025 ([Statistics Norway, 2025](#)), placing Norway well above the EU average of 20 percent; the Nordic countries as a group lead European enterprise adoption ([Bick et al., 2026](#)). Internet access (99 percent) and OECD top-five broadband density provide the infrastructure for rapid diffusion ([OECD, 2024](#)). If AI-related labor market

effects are present, a high-adoption economy is where they would appear first.

2 Related Literature

2.1 Early Empirical Evidence

A small but growing body of work investigates employment shifts associated with generative AI exposure. We review the most closely related studies to ours before turning to null findings and to the measurement of AI exposure.

[Brynjolfsson et al. \(2025\)](#) provide large-scale evidence from the United States. Using private payroll records from ADP covering 3.5 to 5 million workers per month between January 2021 and September 2025, they estimate a Poisson regression event study with firm-by-quintile and firm-by-time fixed effects. Workers aged 22–25 in the most AI-exposed occupational quintile experienced a decline of approximately 15 log points in employment relative to the least-exposed quintile. The effect is concentrated in occupations exposed to automation, as classified by the [Handa et al. \(2025\)](#) decomposition. Occupations exposed primarily to augmentation show no consistent pattern. Wages exhibit little divergence over the same period, consistent with downward nominal rigidity.

[Lodefalk et al. \(2026\)](#) replicate this design in Sweden using employer-employee data covering January 2019 through June 2025. Their difference-in-differences specification includes employer-by-quartile and employer-by-month fixed effects, with AI exposure measured by the DAIOE generative AI index of [Engberg et al. \(2024\)](#) at the four-digit SSYK level. Workers aged 22–25 in the most-exposed quartile show a 5.5 percent employment decline by the first half of 2025. The age gradient is monotonic: the youngest workers decline most, while the oldest (50 and above) gain 1.3 percent. A separate analysis of job postings from Platsbanken attributes part of the decline in vacancies to monetary tightening following the Riksbank rate hike of April 2022, rather than to AI. Pre-trend joint tests reject the null of parallel trends, but Lodefalk et al. argue that the pre-trend slope for the 22–25 group is essentially noise and that any linear extrapolation would bias the estimate toward zero rather than away from it.

[Teeselink \(2025\)](#) uses Revelio Labs data from the United Kingdom and finds that a one standard deviation increase in LLM exposure reduces employment by 0.3 percent. The effects are concentrated in junior positions and in high-wage occupations. [Teeselink and Carey \(2026\)](#) analyze job posting data from 39 countries and find that AI exposure reduces job postings on average. Countries with stricter employment protection exhibit larger hiring reductions, consistent with firms adjusting on the hiring margin when separation

costs are high.

Reported magnitudes vary across countries and specifications: Brynjolfsson et al. report a 16 log-point relative decline in the US, Lodefalk et al. a 5.5 percent decline in Sweden, and Teeselink a 0.3 percent decline per standard deviation in the UK. These figures use different units, baselines, and specifications and should not be read as directly comparable magnitudes.

2.2 Null and Mixed Results

Not all studies find displacement. Humlum and Vestergaard (2026) study Denmark using population-wide register data combined with two survey rounds of approximately 25,000 responses each. Their difference-in-differences design compares workers who adopted AI chatbots to non-adopters. They find a precise null on earnings and no employment effects; at the workplace level, total hours at encouraged-use workplaces are precise enough to rule out effects larger than 2 percent. However, adopters who also switched occupations on average moved to occupations with higher AI exposure and experienced faster earnings growth. This represents an increase in full-time-equivalent occupational mobility, transitioning into higher-wage, higher-AI-exposure occupations.

In their Appendix D.2, Humlum and Vestergaard (2026) use the same 22–25 early-career age definition as Brynjolfsson et al. (2025) and document aggregate declines in early-career employment in several of their exposed occupations in Denmark, including software development, customer support, legal professions, and marketing. The Danish aggregate pattern thus mirrors the US pattern. Their firm-level difference-in-differences then shows that these declines are not driven by firms adopting AI chatbots: estimates rule out effects larger than one third of a percentage point on the early-career employment share at adopting workplaces. They suggest that changes in the macroeconomic environment or anticipatory hiring freezes unrelated to actual adoption may be responsible.

Finnish evidence reinforces the null findings. Kauhanen and Rouvinen (2026) use population-level register data and replicate the Brynjolfsson et al. (2025) methodology. They find no systematic displacement effects and attribute observed employment trends to demographic shifts rather than AI exposure. Earlier Finnish analyses by Kauhanen and Rouvinen (2024, 2025) using a C-AIOE exposure measure likewise found null effects.

Measurement, institutional context, and adoption timing differ across these studies. Humlum and Vestergaard (2026) identify adopters through surveys; Brynjolfsson et al. (2025) and Lodefalk et al. (2026) rely on occupation-level exposure indices. If AI dis-

places through reduced firm entry and exit, individual-level adoption designs may miss the channel. Nordic labor markets share strong employment protection and centralized bargaining that may buffer or delay displacement.

2.3 AI Exposure Measurement

Empirical studies of AI and employment use occupation-level measures that map AI capabilities to task content. The task framework underlying much of this work is set out in ([Acemoglu and Autor, 2011](#); [?; ?](#)). The measures in current use differ in whether they capture potential task exposure, revealed usage, or historical AI progress on the abilities a job requires.

[Eloundou et al. \(2024\)](#) develop the most widely used framework for LLM exposure. Building on the O*NET task database, they employ human annotators and GPT-4 to rate each task on whether access to an LLM (or LLM-powered software) would reduce completion time by at least 50 percent. The resulting measures distinguish between whether this reduction of at least 50 percent would occur with direct LLM exposure or with complementary software. (E_1) measures the share of an occupation's that are directly exposed, while (E_2) measures the share of tasks exposed through LLM and complementary software. Higher-wage occupations score higher on exposure, reversing the pattern observed with earlier waves of automation. The GPT-4 "beta" score, defined as $E_1 + 0.5 \times E_2$, has become a standard measure in subsequent empirical work.

[Handa et al. \(2025\)](#), or the Anthropic Economic Index, measures exposure through data on actual usage. They use a privacy-preserving classifier to map approximately four million Claude.ai conversations to O*NET tasks, dividing each task's conversation count by the number of occupations it is associated with before aggregating to the occupation level. They decompose usage into automation (directive task delegation and feedback-loop conversations with minimal human involvement) and augmentation (collaborative task iteration, learning, and validation), reporting 43 percent automation and 57 percent augmentation across all conversations. This decomposition is important because the two modes may have opposite labor market implications: automation substitutes for worker effort, while augmentation complements it. AI use is highly concentrated by occupation: only about 4 percent of occupations show usage in at least three-quarters of their tasks and about 36 percent in at least a quarter, with Computer and Mathematical occupations alone accounting for 37 percent of all observed conversations.

[Felten et al. \(2021\)](#) take a different approach, linking AI application benchmarks across ten categories (including language modeling, image recognition, and speech recognition)

to the occupational abilities listed in O*NET. Their AI Occupational Exposure (AIOE) score is a weighted sum of AI progress on the abilities required by each occupation. This measure has been extended to a dynamic version (DAIOE) that tracks AI capability growth over time. [Engberg et al. \(2024\)](#) construct a DAIOE variant emphasizing generative AI applications, which is used by [Lodefalk et al. \(2026\)](#) in the Swedish analysis described above.

A deeper conceptual concern is that standard exposure indices aggregate task-level scores using linear formulas, implicitly treating tasks as separable. [Gans and Goldfarb \(2025\)](#) argue that in many occupations, tasks are quality complements rather than independent: output is multiplicative across tasks, as in an O-ring production function ([Kremer, 1993](#)). Under task complementarity, automating some tasks frees the worker to focus on the remaining ones, raising their quality and potentially increasing labor income. The relevant object is then not average task exposure but the structure of bottleneck tasks and how automation reshapes worker time around them. [Imas and Shukla \(2026\)](#) extend this logic, arguing that job dimensionality (the number of distinct tasks in a job) determines whether AI augments or displaces. Low-dimensionality jobs built around a small number of core tasks face higher displacement risk even if their average exposure score is low, while high-dimensionality jobs with many complementary tasks are more likely to be augmented. Exposure indices that average across tasks will overstate displacement risk for complex jobs and understate it for narrow ones.

For Norway specifically, existing evidence documents growing AI adoption across the economy. [Kompetansebehovsutvalget \(2026\)](#) report that AI-related job postings nearly doubled between 2023 and 2024, concentrated in IT but spreading to other sectors.

3 Data, Measures, and Approach

3.1 Norwegian Labor Market Data

Our labor market data come from the *A-ordningen*, Norway’s coordinated employer reporting system. Every Norwegian employer submits a monthly report of all employment relationships, wages, and tax withholdings. The Norwegian Tax Administration administers the system on behalf of Statistics Norway and the Norwegian Labour and Welfare Administration. Statistics Norway compiles the reports into the ARBLONN register, which covers the universe of formal employment in Norway except self-employment, approximately 3.1 million employment records per month (?). We access the data through [microdata.no](#), Statistics Norway’s remote analysis platform, which provides program-

matic access to population-level registers and exports aggregated cell-level tables.

Reference period and inclusion rules. ARBLONN is a monthly status register. The reference period is the week containing the 16th of each month, almost always week 3 (?). On microdata.no, every monthly observation is stamped to the 16th (YYYY-MM-16). An employment relationship is included in month t if its start date is on or before the last day of the reference week and either no end date is recorded or the end date is on or after the first day of the reference week. Two further rules apply. First, when cash earnings are reported for the calendar month but cannot be linked to a contemporaneous or prior-month employment relationship, Statistics Norway constructs a fictional employment relationship and classifies the worker as employed. Second, when wage is paid in the month after an employment relationship ends, the relationship is retained for the last month in which it covered the reference week. As a result, monthly counts cover slightly more workers than a strict snapshot on the 16th would.

Variables. The unit of observation is a person–firm pair in a given month. Statistics Norway aggregates wage components reported by the same employer for the same person into a single person–firm record (?). Table 1 lists the variables we use; all are measured at the 16th of the month.

Table 1: ARBLONN variables used in the analysis

Variable	Description
Occupation code	Four-digit STYRK-08 code for the employment relationship; identical to the International Standard Classification of Occupations (ISCO-08) at the four-digit level (Statistics Norway, 2011).
Cash earnings	Total cash compensation paid by the employer during the calendar month, gross of tax; includes base salary, fixed and irregular supplements, bonuses, overtime pay, and severance.
Hourly wage	Hourly wage reported by the employer.
Contracted position percentage	Contracted hours as a share of full-time, with 100 representing full-time employment.
Overtime hours	Overtime hours reported by the employer for the calendar month.
Start date	Contractual start date of the employment relationship active on the 16th. We define a new-job indicator equal to one if the start date falls within the 30 days ending on the 16th of month t .
Institutional sector	Four-digit code from Statistics Norway’s Business Register, collapsed into state (codes 1110, 1120, 6100), municipal (codes 1510, 1520, 6500), and private (all other codes).

Sample. Age is computed at monthly precision by subtracting birth year from calendar year and decrementing by one if the birth month exceeds the current month. Workers are assigned to four decade age groups: 21–30 (early career), 31–40, 41–50, and 51–60 (senior). We exclude individuals under age 21, whose labor market attachment is dominated by education transitions, and those over 60, who are affected by retirement timing. The sample period is January 2021 through February 2026.

3.2 Institutional Context

Several features of the Norwegian labor market are relevant for interpreting AI exposure patterns. Norway’s public sector employs approximately 30 percent of the workforce, split between state and municipal employers. Wage determination is centralized: unions and employer organizations negotiate sector-level agreements that compress the wage distribution relative to market-based systems such as the United States. Employment protection legislation is strong by international standards, with notice periods and restrictions on dismissal that slow labor market reallocation.

The period we study coincides with a monetary policy tightening: Norges Bank raised the policy rate from zero percent in September 2021 to 4.5 percent by January 2024 (Figure A1). This tightening affected labor demand broadly, and any AI-specific employment patterns need to be read against this macroeconomic background.

A demographic factor also affects the youngest age group. The early-career cohort shrank slightly between 2022-Q1 and 2025-Q1 (Figure A2). Although small, this trend works in the same direction as any AI-related decline in young employment, and should be kept in mind when reading the indexed series.

A small set of Norwegian reports document AI adoption and its early labor-market correlates. Jordell et al. (2026) surveys 4,294 firms and reports that the share using AI rises from 24 percent in 2023 to 55 percent in 2025, concentrated in ICT, finance, and professional services. Vatne et al. (2026) regresses occupation-level unemployment growth on a Claude-based exposure measure for 2022–2025 and reports a positive gradient and no wage relationship, without decomposition by age or sector. Flobakk-Sitter et al. (2024) interviews members of public-sector unions and finds AI use that is exploratory and concentrated in augmentation rather than automation. Riksrevisjonen (2024) reports that more than half of Norwegian state agencies have no experience with AI tools, and Arbeids- og velferdsdirektoratet (2025) reviews the broader labor-market outlook without occupation-level adoption data.

3.3 AI Exposure Measures

Our main measure of occupational AI exposure is the Eloundou et al. (2024) GPT-4 exposure score. As a complement, the appendix uses the Handa et al. (2025) automation and augmentation shares, which capture revealed patterns of AI usage. Table 3 summarizes these measures.

Eloundou et al. (2024), GPT-4 exposure. Eloundou et al. (2024) combine expert and GPT-4 assessments of each occupation’s tasks to estimate the share of tasks for which access to a large language model (LLM) would reduce completion time by at least 50 percent. The occupation-level score, denoted β , is defined as $E_1 + 0.5 \times E_2$, where E_1 is the share of tasks directly exposed to an LLM and E_2 is the share exposed when complementary software is included. The measure captures theoretical LLM capability as assessed in early 2023. We map 397 STYRK-08 codes, covering 97.5 percent of all four-digit occupations in our data.

Handa et al. (2025), Anthropic Economic Index. Handa et al. (2025) measure revealed AI usage patterns from a sample of Claude.ai conversations, with each conversation classified by O*NET task. The occupation-level score reflects the share of conversations associated with that occupation’s tasks, decomposed into automation and augmentation modes as described in Section 2. We map 352 STYRK-08 codes (86.5 percent coverage). The lower coverage reflects the concentration of observed Claude usage: the top seven task categories account for 22 percent of all conversations, leaving many occupations with negligible observed usage.

Crosswalk Construction

Both measures originate in US SOC codes. The Handa et al. measure uses SOC 2010 codes directly. The Eloundou et al. measure uses SOC 2018 codes, which we first map to SOC 2010 using the official BLS crosswalk (November 2017). From SOC 2010, we apply the BLS SOC-to-ISCO crosswalk (August 2012, updated June 2015) to obtain ISCO-08 codes. Since STYRK-08 is identical to ISCO-08 at the 4-digit level (Statistics Norway, 2011), this final step is an identity mapping.

The BLS crosswalk contains partial matches: 38.8 percent of its rows are flagged as one-to-many or many-to-one mappings. When multiple SOC codes map to a single STYRK-08 code, we take the unweighted average of their exposure scores. We compute quality flags for each STYRK-08 code, including a partial-match indicator and the maximum fan-

out, defined as the number of SOC codes contributing to a single STYRK-08 score. Of the 397 STYRK-08 codes with an Eloundou et al. score, 57.7 percent have at least one partial-match contributor.

3.4 Empirical Approach

We construct normalized time series following the approach in Brynjolfsson et al. (2025), Figures 1–3. Each series is indexed to October 2022 = 1, the month immediately preceding ChatGPT’s public release on November 30, 2022. For outcome y in cell c at month t , the normalized series is $\tilde{y}_{c,t} = y_{c,t}/y_{c,\text{Oct 2022}}$. Changes after October 2022 are expressed as proportional deviations from that pre-ChatGPT level.

For each AI exposure measure, we rank all four-digit STYRK-08 occupation codes by their exposure score and assign them to quintiles, from quintile 1 (least exposed) to quintile 5 (most exposed). Quintiles are formed on the occupation distribution (each four-digit code counts once, regardless of its employment share) rather than on the employment-weighted distribution. Quintile assignment is performed separately for each measure, so an occupation may fall in different quintiles under different measures. We then aggregate employment counts and other outcomes within each quintile-by-age-group cell and construct the normalized time series.

The primary outcome is employment indexed to October 2022, by decade age group and exposure quintile. Headline results are for the private sector, where AI-exposed occupations cluster; the public sector is reported in the appendix. Secondary outcomes, available from 2021 onward, include monthly earnings, hourly wages, contracted position percentage, overtime hours, and the share of new jobs.

Data Availability

The AI exposure measures and crosswalk files used in this paper are publicly available. The Eloundou et al. (2024) GPT-4 exposure scores are available at <https://github.com/openai/GPTs-are-GPTs> (the file `data/occ_level1.csv`, supplementary data from the *Science* article). The Handa et al. (2025) Anthropic Economic Index and the Anthropic (2026) `job_exposure` measure are available at <https://github.com/anthropics/anthropic-economic-index>. The Felten et al. (2021) AIOE data are available as an appendix to the *Strategic Management Journal* article. The BLS SOC-to-ISCO crosswalk is available at <https://www.bls.gov/soc/soccrosswalks.htm>. The STYRK-08 classification is documented by Statistics Norway at <https://www.ssb.no/klass/klassifikasjoner/7>. Norwegian labor market

Table 2: Five largest occupations by Eloundou exposure quintile, April 2025.

Code	Occupation	Eloundou score	Employment (Apr 2025)	Share of quintile
<i>Quintile 1 (least exposed)</i>				
5321	Helsefagarbeidere	0.113	115,058	22.3%
9112	Renholdere i bedrifter	0.028	69,745	13.5%
7411	Elektrikere	0.128	31,310	6.1%
8160	Operatører innen næringsmiddelproduksjon	0.093	26,176	5.1%
5153	Vaktmestre	0.000	24,041	4.7%
<i>Quintile 2</i>				
5311	Barnehage- og skolefritidsassistenter mv.	0.249	110,807	22.1%
5329	Andre pleiemedarbeidere	0.216	77,616	15.5%
7115	Tømrere og snekkere	0.144	42,650	8.5%
2342	Førskole-/barnehagelærere	0.230	35,248	7.0%
5322	Hjemmehjelper	0.219	22,950	4.6%
<i>Quintile 3</i>				
5223	Butikkmedarbeidere	0.342	131,014	16.8%
2341	Grunnskolelærere	0.329	88,502	11.3%
2223	Sykepleiere	0.343	67,146	8.6%
4321	Lagermedarbeidere og material-forvaltere	0.325	35,442	4.5%
2221	Spesialsykepleiere	0.343	32,390	4.2%
<i>Quintile 4</i>				
1120	Administrerende direktører	0.494	40,124	7.6%
2310	Universitets- og høyskolelektorer/-lærere	0.486	39,127	7.4%
1420	Varehandelssjefer	0.481	35,076	6.6%
2142	Sivilingeniører (bygg og anlegg)	0.482	27,706	5.2%
1219	Andre administrative ledere	0.492	27,529	5.2%
<i>Quintile 5 (most exposed)</i>				
2422	Høyere saksbehandlere i offentlig og privat virksomhet	0.566	91,269	16.9%
4110	Kontomedarbeidere	0.595	59,600	11.1%
3322	Selgere (engros)	0.572	44,698	8.3%
2511	Systemanalytikere/-arkitekter	0.626	32,939	6.1%
2519	Andre programvare- og applikasjonsutviklere	0.765	22,143	4.1%

data are accessed through the microdata.no platform (<https://microdata.no>). The aggregated tabulations underlying our analysis are available from the authors upon request.

Table 3: AI Exposure Measures Mapped to Norwegian STYRK-08 Occupations

Measure	Type	Vintage	STYRK-08 coverage	Conceptual basis
Eloundou et al. (2024) β	Theoretical	2023	397 / 407 (97.5%)	Expert and GPT-4 assessment of task exposure to LLMs; $\beta = E_1 + 0.5 \times E_2$
Handa et al. (2025) overall	Revealed usage	2025	352 / 407 (86.5%)	Share of Claude conversations per O*NET task
Felten et al. (2021) AIOE	Ability-based	2018–2021	392 / 407 (96.3%)	AI application performance linked to occupational abilities via O*NET
Anthropic (2026) job exposure	Observed, time-weighted	2026	388 / 407 (95.3%)	Time-weighted Claude usage with automation penalty

Notes: All measures are mapped to STYRK-08 via a SOC 2010 $\rightarrow$ ISCO-08 crosswalk (BLS, 2012); the Eloundou et al. measure additionally uses the BLS SOC 2018 $\rightarrow$ SOC 2010 crosswalk (November 2017) as a first step. STYRK-08 is identical to ISCO-08 at the four-digit level (SSB Notater 17/2011). When multiple SOC codes map to a single STYRK-08 code, the STYRK-08 score is the unweighted mean of the contributing SOC scores. Coverage is expressed as the share of 407 STYRK-08 four-digit codes observed in our employment data.

4 Results

4.1 Employment in Selected Occupations

Figure 1 displays normalized employment per capita for four occupation groups with high theoretical AI exposure: software developers, ICT systems analysts, customer service agents, and ICT operations technicians.¹ Each panel plots employment per capita indexed to October 2022 = 1, separately by age group, with a vertical line marking the public release of ChatGPT on November 30, 2022.

The gradient is largest in software development. The 22–25 series falls from 1.0 at the index date to about 0.70 by late 2025; the 35–40 series rises to 1.15 over the same window. The 26–30 series also falls, less steeply. ICT systems analysts look the same: 22–25 down to 0.75, 35–40 up to 1.10.

Customer service agents show the same age gradient, with more month-to-month variation. The 22–25 group fluctuates around 0.95 before declining toward the end of the

¹ISCO-08 codes: software developers 2512–2514 and 2519; ICT systems analysts 2511; customer service agents 4222; ICT operations technicians 3511.

period, while the 31–40 groups grow to approximately 1.10–1.15. ICT operations technicians exhibit a weaker version of the pattern: the youngest group declines modestly to around 0.90, and the age separation is less pronounced than in the other three panels.

All four occupations rank high on AI exposure indices and feature in similar case studies in [Brynjolfsson et al. \(2025\)](#); in all four the youngest workers lose ground while older workers do not. Section 4.2 incorporates all occupations systematically.

4.2 Employment by AI Exposure Quintile

The case studies above examine occupations chosen for their AI relevance. A more systematic approach ranks all occupations by their AI exposure score and examines whether high-exposure occupations exhibit different employment trends than low-exposure occupations. We assign each occupation to a quintile based on its Eloundou et al. GPT-4 exposure score, from Q1 (least exposed) to Q5 (most exposed), and plot normalized employment by quintile within each age group.

Eloundou et al. Exposure

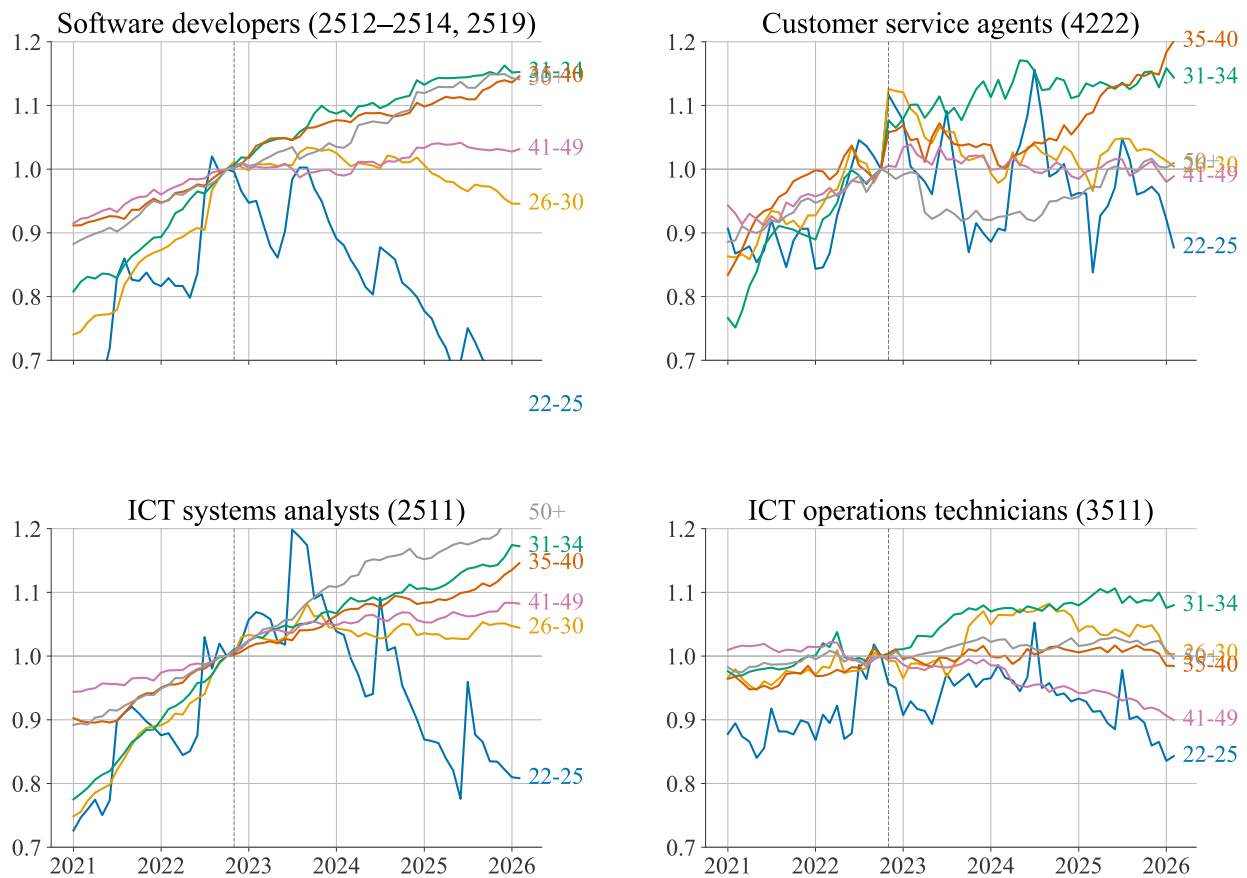
Figure 2 plots quintile-by-age employment for the private sector, indexed to October 2022. Each of the four panels is one decade age group, with quintile lines running from light blue (Q1, least exposed) to dark blue (Q5, most exposed) and a red line for the overall age-group trend.

The most-exposed occupations do not lose ground among the youngest workers. In the 21–30 and 31–40 groups the Q5 series tracks or runs above Q1 after the ChatGPT release. The negative gradient appears instead at ages 41–50, where Q5 ends about four points below Q1 by early 2026. The 51–60 group shows no clear separation.

Table ?? reports the corresponding difference-in-differences estimates: the average post-October-2022 change in each quintile relative to Q1, by age group, for employment, new hires, and log monthly earnings. The employment estimates are positive and significant for the most-exposed quintiles at ages 21–40 (Q5 about +0.07 log points in both groups) and negative at 41–50 (Q5 -0.053 , $p < 0.05$). New hires fall for the most-exposed quintile at 41–50 (Q5 -0.084). Log earnings show no systematic exposure gradient.

The companion automation and augmentation quintile plots based on the Handa et al. decomposition are reported in Appendix Figures A5 and A6.

Employment per capita by age group (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1), normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). STYRK-08 occupation codes in parentheses. Age is calculated using month of birth.

Figure 1: Employment per capita (October 2022 = 1) by age group for selected high-exposure occupations. Employment counts divided by population in each age group (Statistics Norway table 07459) with composition adjustment (reference: 2021-Q1). Panels show software developers (ISCO-08 2512–2514, 2519), ICT systems analysts (2511), customer service agents (4222), and ICT operations technicians (3511). The vertical dashed line marks November 30, 2022 (ChatGPT release).

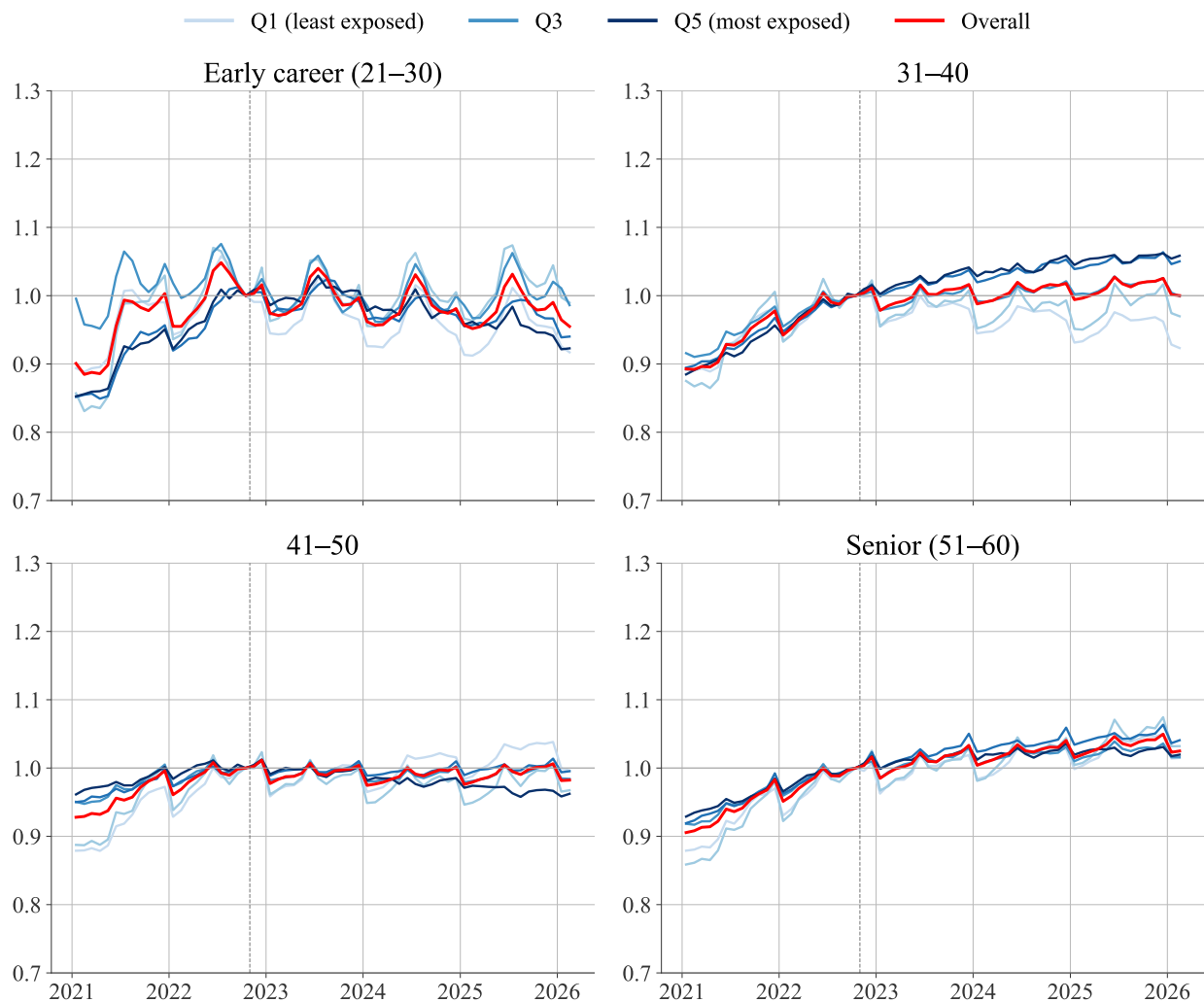


Figure 2: Employment (October 2022 = 1) by Eloundou et al. GPT-4 exposure quintile and decade age group, private sector. Q1 (lightest) = least exposed; Q5 (darkest) = most exposed. Red line = overall age-group trend. Vertical dashed line marks ChatGPT release.

Table 4: Cell-level difference-in-differences by AI exposure quintile, private sector. Each entry is the post-October-2022 change in the indicated quintile relative to Q1 (least exposed), by decade age group. Employment and new hires are Poisson; log monthly earnings is OLS weighted by cell employment. Occupation and month fixed effects; standard errors clustered by occupation. Sample 2021m1–2025m4. *, **, ***: $p < 0.1, 0.05, 0.01$.

	Early career (21–30) (1)	31–40 (2)	41–50 (3)	Senior (51–60) (4)
<i>Panel A. Employment (Poisson)</i>				
Q2 × Post	0.0469 (0.0355)	0.0128 (0.0353)	-0.0283 (0.0301)	0.0112 (0.0251)
Q3 × Post	-0.0073 (0.0400)	0.0213 (0.0193)	-0.0416** (0.0189)	-0.0118 (0.0166)
Q4 × Post	0.0674** (0.0310)	0.0614*** (0.0205)	-0.0344* (0.0188)	0.0007 (0.0172)
Q5 × Post	0.0700** (0.0276)	0.0695*** (0.0217)	-0.0528*** (0.0198)	-0.0127 (0.0208)
<i>Panel B. New hires (Poisson)</i>				
Q2 × Post	0.0649 (0.0413)	0.0579 (0.0575)	0.0014 (0.0474)	-0.0133 (0.0501)
Q3 × Post	0.0539 (0.0389)	0.0833* (0.0490)	0.0216 (0.0440)	0.0309 (0.0508)
Q4 × Post	0.0032 (0.0351)	0.0482 (0.0414)	-0.0244 (0.0329)	0.0301 (0.0343)
Q5 × Post	-0.0201 (0.0383)	-0.0084 (0.0439)	-0.0841** (0.0341)	-0.0042 (0.0339)
<i>Panel C. Log monthly earnings (OLS)</i>				
Q2 × Post	-0.0104 (0.0074)	-0.0027 (0.0053)	-0.0055 (0.0060)	0.0010 (0.0052)
Q3 × Post	-0.0364*** (0.0127)	-0.0094 (0.0060)	-0.0078 (0.0074)	0.0012 (0.0050)
Q4 × Post	-0.0037 (0.0075)	0.0018 (0.0034)	-0.0006 (0.0042)	0.0120*** (0.0043)
Q5 × Post	-0.0027 (0.0051)	0.0071* (0.0036)	0.0070* (0.0039)	0.0161*** (0.0039)
Occupations	393	392	392	391
Observations (count)	20,436	20,384	20,384	20,332
Observations (wage)	19,326	19,414	19,376	19,296

4.3 Automation versus Augmentation

We follow [Brynjolfsson et al. \(2025\)](#) on the automation-vs-augmentation split. Ranking the youngest workers by Handa automation share gives a modest negative gradient (Appendix Figure A5); ranking them by augmentation share gives no gradient at all (Appendix Figure A6). The split points toward displacement rather than augmentation, but the Handa classification only captures conversation mode, not what workers actually do on the job. An occupation scored as “automation-exposed” through conversation mode may in practice use AI for augmentation.

4.4 Sector Heterogeneity

Figure 5 decomposes the employment quintile analysis by institutional sector. The three panels show the private sector, the municipal sector, and the state sector, with Eloundou et al. quintile lines plotted for each.

When all age groups are pooled, no quintile gradient is visible in any of the three sectors: the affected age groups are a small share of total employment, so the age-specific gradient documented in Figure 2 washes out in the pooled series. When broken down by sector (Appendix Figures A19–A21), the age-exposure gradient reappears in the private sector but is absent in the municipal and state sectors.

The public-sector null could reflect slower adoption, hiring insulated from market conditions, stronger employment protection, or a different occupational mix; we cannot separate these from the aggregate data. The adoption channel does have some support: [Riksrevisjonen \(2024\)](#) report that more than half of Norwegian state agencies have no experience with AI tools, so theoretical exposure in the public sector does not translate into implementation. The public sector is dominated by health, education, and administration; private high-exposure occupations cluster in IT, finance, and business services.

4.5 Wages and Other Outcomes

Figure 6 displays monthly cash earnings by Eloundou et al. exposure quintile and age group. Strong seasonal patterns dominate the wage plots: December earnings spike to 1.4–1.5 times the baseline due to Christmas bonuses, and June earnings are elevated due to statutory holiday pay. [Brynjolfsson et al. \(2025\)](#) and [Humlum and Vestergaard \(2026\)](#) find no wage divergence over a similar horizon in the US and Denmark. Norway’s two-tier bargaining system ([Bhuller et al., 2022](#)) with sectoral wage floors and strong employment protection make the quantity margin the more likely site of adjustment.

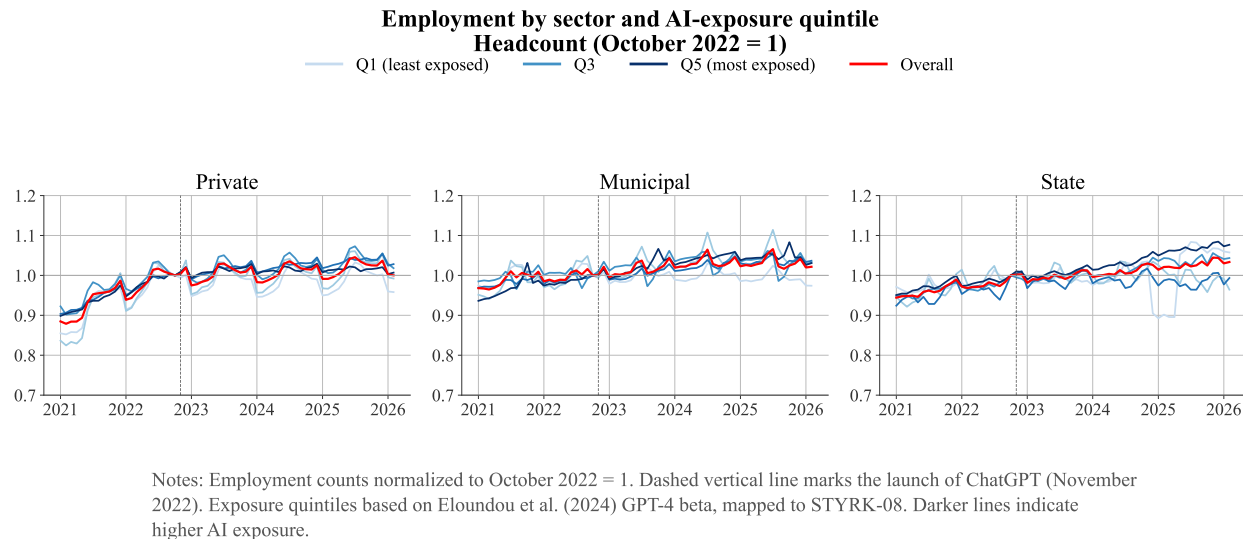


Figure 3: Employment by Eloundou et al. exposure quintile, decomposed into private, municipal, and state sectors.

Four additional outcomes are reported in the appendix: hourly wages (Figure A13), contracted position percentage (Figure A12), overtime hours (Figure A14), and the share of new jobs (Figure A15). None shows a differential trend by exposure quintile, consistent with the extensive margin as the candidate site of adjustment.

4.6 Poisson Regression Estimates and Within-Firm Reallocation

The quintile series so far normalize employment by the population in each age group, the right denominator for asking whether cohort-level employment shifts away from AI-exposed occupations. We turn now to the worker counts and ask if they reallocate across occupations? Following Brynjolfsson et al. (2025), we estimate Poisson event studies on the counts.

We report two specifications. The first uses the cell-level aggregates that underlie the rest of Section 4; it covers the universe of formal employment and includes any composition shifts *between* firms. The second uses individual-level employer-reported records, restricted to private-sector firms with at least 20 workers in the age window over the full 2021m1–2025m7 panel, and adds firm fixed effects to remove between-firm composition. The two are not nested: they differ in sample (universe vs. panel-balanced private-sector firms above the size threshold), in identifying variation (within-occupation time variation vs. within-firm-by-quintile time variation), in implicit Poisson weighting (by occupation size vs. by firm-by-quintile cell size), and in clustering unit (occupation vs. firm). Differ-

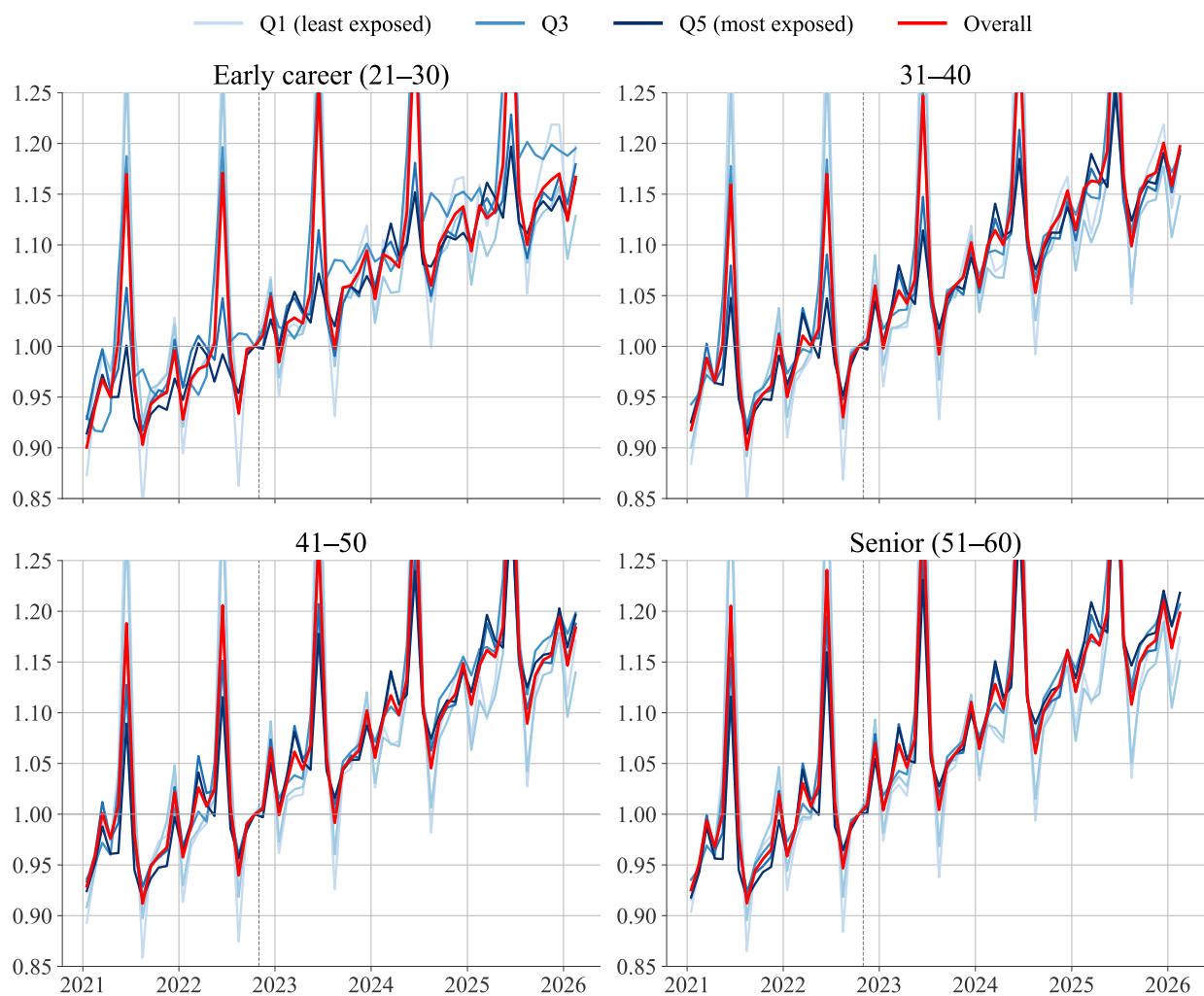


Figure 4: Monthly cash earnings (October 2022 = 1) by Eloundou et al. exposure quintile and decade age group, private sector. Strong seasonal patterns (December bonuses, June holiday pay) are visible in all panels.

ences between them therefore to some extent mix between-firm reallocation with sample selection and the choice of FE level. We discuss both below.

Cell-level. The cell-level microdata are tabulated at the (occupation $\times$ age bin $\times$ month) level. We estimate

$$\log E[\text{count}_{j,a,t}] = \alpha_j + \beta_t + \gamma_{q,k},$$

separately per age bin a , where j indexes occupations and $q(j)$ is the Eloundou GPT-4 β quintile of occupation j . The occupation FE absorbs the permanent employment level of each occupation; the month FE absorbs the aggregate time path of the age cohort. Standard errors are clustered at occupation. Reference: Q1 and $k = -1$ (October 2022).

Figure 7 reports the resulting $\gamma_{q,k}$ for $q \in \{2, 3, 4, 5\}$ across the six age bins, in log points. The same picture emerges as in the linear rate figure (Figure 2). At 22–25, Q5 reaches -15 to -20 log points by 2025, Q3 and Q4 fall less steeply, and Q2 is flat to slightly negative; the ordering is monotone. The 26–30 panel has the same sign at smaller magnitude. From age 31 upward, the gradient flattens or reverses, with Q5 growing slightly above Q1 in the mid-career bins.

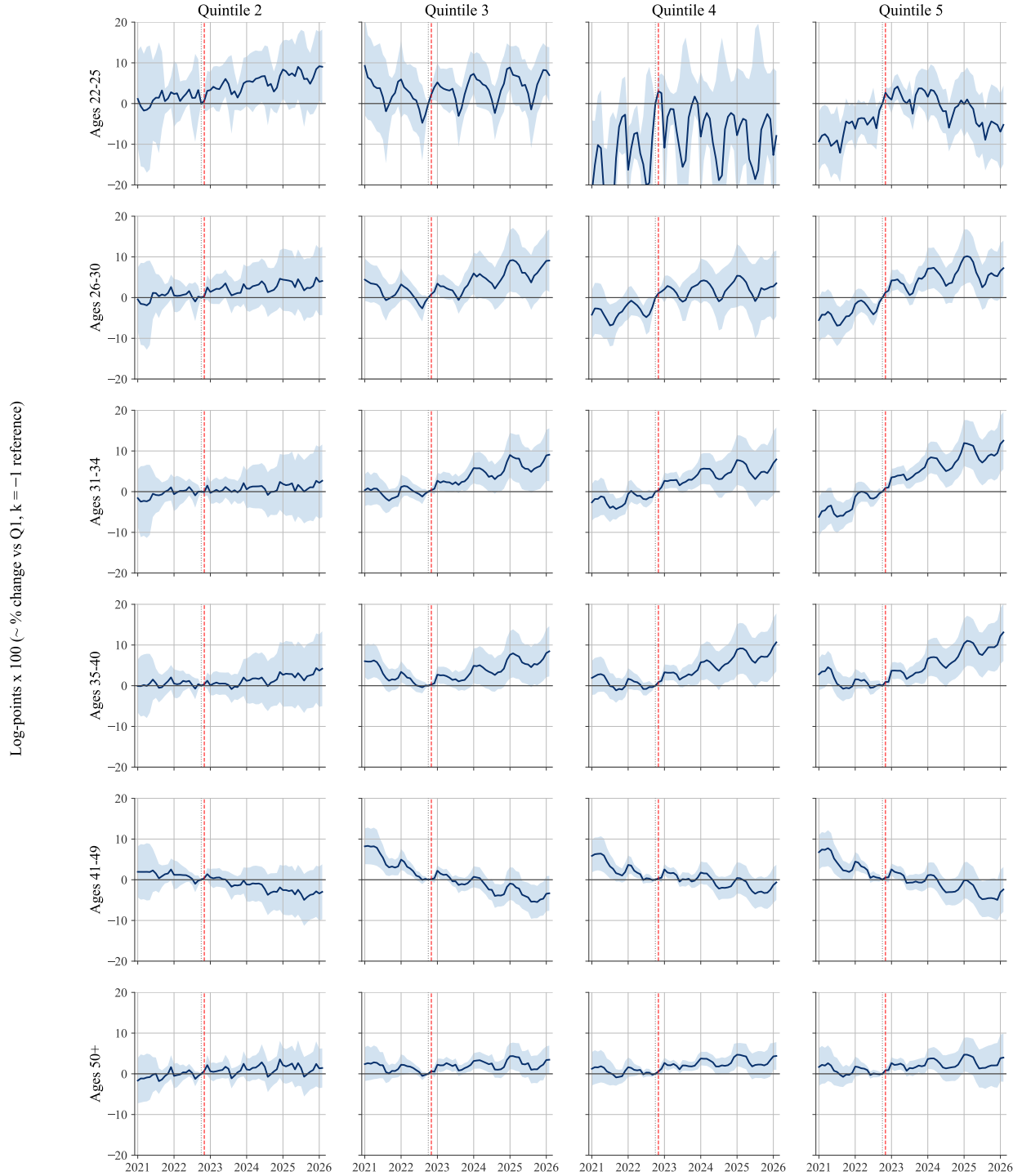
Within-firm. The individual-level records are available to us only through July 2025, six months short of the cell-level series we use for the main analysis. For that reason we do not adopt the within-firm specification as our headline design. We run it here as a check on the aggregated results: if the cohort-level pattern documented above were driven by reorganization *within* continuing firms, that should show up in a within-firm composition specification à la Brynjolfsson et al. (2025).

The panel covers private-sector firms from 2021m1 through 2025m7, restricted to ages 22–55 with positive cash earnings and a valid occupation code. We keep firms with at least 20 workers in the age window, fill synthetic zeros for (firm, age bin, occupation) cells ever observed, and rank occupations into quintiles on Eloundou et al. GPT-4 β . The estimating sample is 48.8 million worker-months across 18,400 firms. The unit of observation is the (firm, age bin, quintile, month) cell; the outcome is the count of workers in that cell. A firm is not assigned to a quintile, it typically employs workers across several occupations spanning multiple quintiles, and what firms differ in is the *share* of employment in each quintile. We estimate with firm-by-quintile and firm-by-month fixed effects separately per age bin:

$$\log E[\text{count}_{f,q,a,t}] = \alpha_{f,q} + \beta_{f,t} + \gamma_{q,k},$$

with reference $k = -1$ and standard errors clustered at firm. The firm-by-quintile FE

Employment by AI-exposure quintile vs. Q1, by age cohort **cell-level event study, Poisson**



Sample: cell-level microdata.no employment counts, 22–69-year-olds, 2021m1–2026m2. Source: data/01_occ_agemonth_count_2021_2026.csv. Specification: Poisson GLM with occupation FE (yrke4), month FE, and quintile x event-time interactions (Q1 and k = -1 omitted), estimated separately per age bin. SE clustered at occupation. Aggregates over firms (microdata.no exports collapse the firm dimension), so this is the counterpart to the firm-FE estimator in Appendix C with the firm dimension removed --- the $\gamma_{q,k}$ coefficient combines within-firm reallocation and across-firm scale. Dashed red line: November 2022 (k = 0). Y-axis: $\gamma_{q,k}$ (log-points) x 100, approximately percent change in q employment relative to Q1 from October 2022 to event time k.

Figure 5: Poisson event-study coefficients $\gamma_{q,k}$ for $q \in \{2, \dots, 5\}$ vs. $q = 1$, by age bin. Cell-level microdata.no aggregates (occupation $\times$ age $\times$ month); FE: occupation and month.

absorbs each firm’s permanent employment level in each quintile (so heterogeneity in firms’ occupational mix is netted out); the firm-by-month FE absorbs each firm’s overall employment trajectory. The remaining variation identifies, within a firm and age bin, whether Q5 employment evolves differently from Q1 employment after October 2022.

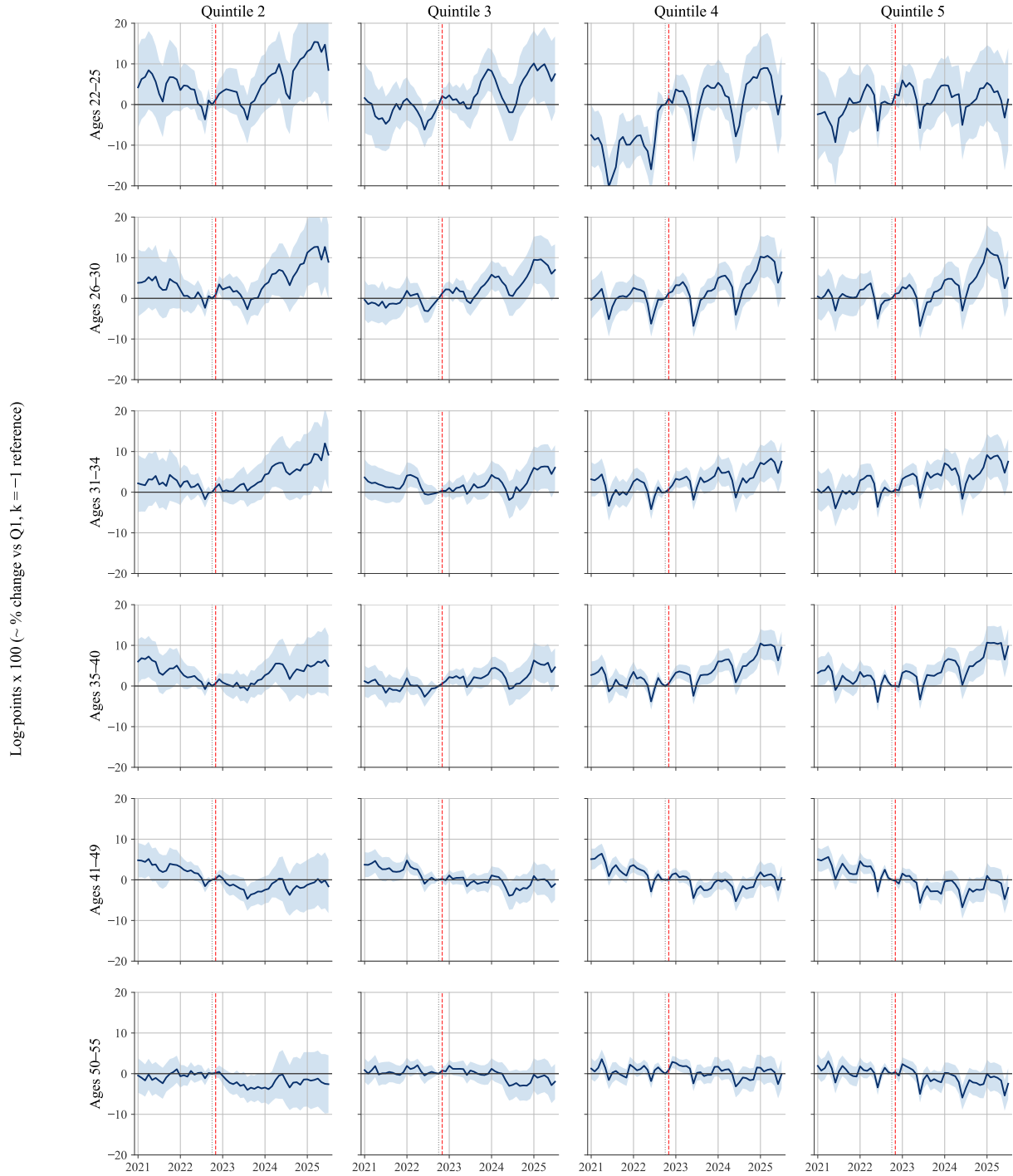
Figure 8 reports the same Q2–Q5 vs. Q1 grid for the firm-FE specification. The 22–25 panel is centered near zero with wide CIs and no break at November 2022; the cell-level grid in Figure 7 reaches -15 to -20 log points in the same panel. Ages 26–30 through 35–40 drift up to $+8$ to $+10$ log points by mid-2025, monotone in exposure. Ages 41–49 and 50–55 are flat to slightly negative.

Interpretation. The cell-level analysis counts every worker in the formal economy and so picks up the cohort-level decline that drives the linear rate figures. The firm-FE analysis sees only the panel of private-sector firms with at least 20 workers in the age window throughout 2021m1–2025m7, and within those firms it estimates how the Q5 vs. Q1 mix of young workers evolves over time. A firm-FE coefficient on Q5 22–25 close to zero therefore says only that the young Q5/Q1 mix is roughly unchanged *within* the firms that meet the panel restriction, it is silent about firms that shrank below the size cutoff, exited, or were never in private employment, and about new firms that entered after 2021. One potential reading of a sharp cell-level decline combined with a near-zero firm-FE coefficient is that the cohort-level fall operates between firms rather than through reorganization inside continuing firms: Q5-exposed firms may be growing more slowly (or shrinking, exiting, failing to be replaced) relative to Q1-exposed firms, while each surviving firm’s internal mix of young workers across exposure quintiles stays roughly the same. The mid-career positive coefficients from the firm-FE specification go further and imply that continuing firms reweight toward exposed occupations among workers in their early thirties to early forties.

Table 5 reports the pooled young $\times$ post $\times$ exposure_std triple-difference: Under the firm-FE Poisson regression, the coefficient is -0.0022 (SE 0.0063); the linear-OLS analogue on the count is -0.0058 (SE 0.0104). Within the same firm and month, young workers in higher-exposure occupations do not decline relative to older ones in the same firm. The cohort-level decline therefore appears to run through the extensive margin (which firms hire young workers, and which young workers find employment at all) rather than through reshuffling of young workers across exposure quintiles inside continuing firms.

Intensive margin. Table 5 brings the extensive-margin pooled triple-differences together with the intensive-margin outcomes. The intensive-margin rows are linear-OLS triple-

Employment by AI-exposure quintile vs. Q1, by age cohort firm-FE event study, Poisson



Sample: private-sector foretak (sekt = 3), 22–55-year-olds, 2021m1–2025m7. Specification: $\text{fepois}(\text{count_all} \sim i(k, \text{ai_q, ref} = -1, \text{ref2} = 1) \mid \text{foretak}^q + \text{foretak}^{\text{month}})$, clustered at foretak, estimated separately per age bin. Reference: October 2022 ($k = -1$) and Q1. Dashed red line: November 2022 (GPT-4 era). Tick marks at January of each year. Shaded band: 95 % CI. Y-axis: $\gamma_{q,k}$ (log-points) x 100, approximately percent change in q employment relative to Q1 from the reference month to event time k . All panels share the y-axis (Poisson log-points are directly comparable across ages and quintiles). Source: [from_secure_server/coefficients/coef_event_study_fepois.csv](#).

Figure 6: Poisson event-study coefficients $\gamma_{q,k}$ for $q \in \{2, \dots, 5\}$ vs. $q = 1$, by age bin. Individual-level employer-reported records, private-sector firms, ages 22–55, 2021m1–2025m7; FE: firm $\times$ quintile and firm $\times$ month. Shaded band: 95% CI clustered at firm. Reference: October 2022. Dashed red line: November 2022.

differences with firm-by-age, firm-by-month, and age-by-month fixed effects, weighted by cell employment. The reported coefficient is on $\text{young} \times \text{post} \times \text{exposure_std}$: the differential post-October-2022 response for 22–25 year-olds relative to older workers in the same firm and month, per one standard deviation of occupational exposure. After October 2022, a one standard deviation increase in exposure is associated with a 0.8 percent lower monthly wage and a 0.4 percentage point lower contracted position percentage for 22–25 year-olds relative to older workers within the same firm. Base hours and overtime are unchanged. The intensive-margin magnitudes are small and would not appear in the quintile-level series of Section 4.5, but they go in the same direction as the extensive-margin findings.

Table 5: Pooled triple-difference estimates, firm-FE specification.

Outcome	Coef.	SE	<i>t</i>
<i>Extensive margin (employment count)</i>			
Headcount, Poisson (log points)	−0.0022	0.0063	−0.34
Headcount, linear OLS	−0.0058	0.0104	−0.55
<i>Intensive margin</i>			
Log monthly wage	−0.0077	0.0022	−3.50
Position pct	−0.396	0.099	−4.02
Log base hours	−0.0027	0.0027	−1.00
Overtime hours	−0.011	0.027	−0.43

Coefficient on $\text{young} \times \text{post} \times \text{exposure_std}$. All rows use $\text{firm} \times \text{age} + \text{firm} \times \text{month} + \text{age} \times \text{month}$ fixed effects. Extensive-margin rows (Poisson and linear OLS on the count) are unweighted; intensive-margin rows are weighted by cell employment so the cell-level OLS reproduces the underlying individual-level regression. SE clustered at firm.

5 Robustness

5.1 Crosswalk Quality

The mapping from US SOC codes to Norwegian occupation codes introduces measurement error. When many SOC codes map to a single Norwegian code, the resulting exposure score is an average over heterogeneous US occupations that may not correspond well to the Norwegian occupation category. The fan-out metric, defined as the number of SOC codes contributing to a single Norwegian code, provides a measure of crosswalk quality. Higher fan-out implies a less precise mapping.

Figure 9 presents the Eloundou et al. quintile-by-age grid under three crosswalk qual-

ity restrictions. The first column uses all 397 mapped codes, replicating the baseline in Figure 2. The second column restricts to the 316 codes with fan-out of five or fewer, retaining approximately 80 percent of total employment. The third column applies the most restrictive filter, retaining only the 251 codes with fan-out of two or fewer, covering approximately 69 percent of employment.

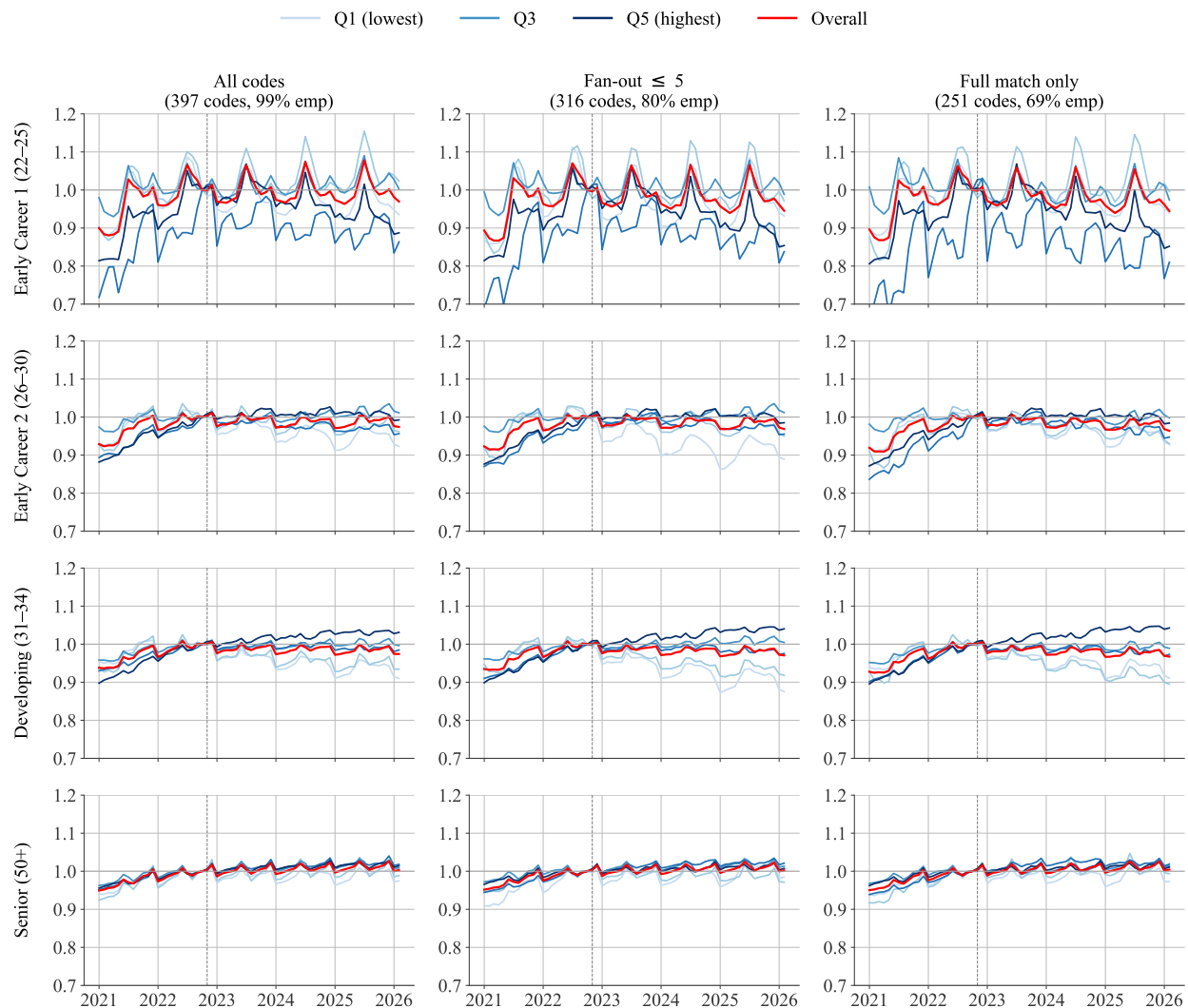
The age-exposure gradient for the 22–25 group persists across all three specifications. Under the most restrictive filter, the series are noisier due to smaller cell sizes, but Q5 continues to decline relative to Q1 after late 2022. For older age groups, none of the three specifications produces a systematic quintile gradient. The gradient therefore persists in higher-quality crosswalk subsamples.

5.2 Population Adjustment

All main figures report employment per capita rather than raw headcounts, dividing employment counts by the resident population in each age group (Statistics Norway table 07459, interpolated from annual to monthly). In addition, a composition adjustment holds the within-group age distribution fixed at its 2021-Q1 level, removing the effect of shifting age composition within broad age bins. The per-capita adjustment removes the mechanical decline from the cohort-size shrinkage noted in Section 3.2. For the core age groups 22–69, the composition adjustment factors are close to unity (within 0.1 percent), indicating that within-group age shifts are negligible over this period. For the youngest (≤ 21) and oldest (70+) groups, the adjustment is more consequential (up to 5–6 percent). Appendix Figures A3 and A4 show the raw headcount versions of Figures 1 and 2 for comparison with Brynjolfsson et al. (2025). The gradient survives unchanged; the per-capita adjustment compresses the age spread (the 22–25 software developer decline is 30 percent per capita versus 35 percent in raw headcounts).

Additional caveats. The Norwegian occupation coding is stable over 2021–2026 by design, but as AI-relevant occupations evolve, employers may shift how they classify ambiguous job titles, introducing a reclassification channel that runs in the opposite direction from the one we are after.

Sensitivity: Eloundou et al. exposure quintiles by crosswalk quality



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1). Sensitivity of Figure 2 to crosswalk quality restrictions. Left: all mapped codes. Center: excluding occupations where any contributing SOC code partially maps to >5 ISCO codes. Right: only full SOC-ISCO matches (fan-out ≤ 2). Quintiles re-assigned within each restricted sample. Eloundou et al. (2024) GPT-4 beta exposure.

Figure 7: Crosswalk quality sensitivity: Eloundou et al. exposure quintiles by age group under three fan-out restrictions. Left column: all 397 Norwegian occupation codes (baseline). Center column: fan-out ≤ 5 (316 codes, approximately 80 percent of employment). Right column: fan-out ≤ 2 (251 codes, approximately 69 percent of employment). All series indexed to October 2022 = 1. Fan-out denotes the number of SOC 2010 codes contributing to a single Norwegian code.

6 Discussion

6.1 Cross-Country Comparison

Table 6 situates the Norwegian patterns within the cross-country evidence reviewed in Section 2. The table separates descriptive time series (Panel A) from formal econometric estimates (Panel B), because several studies employ both approaches and the magnitudes often differ.

All five countries with descriptive analyses show an early-career decline in AI-exposed occupations, with the exception of Finland. The Norwegian per-capita decline of approximately 30 percent for young software developers is between the US (−20%, [Brynjolfsson et al., 2025](#)) and the Swedish raw figure (−44%, [Lodefalk et al., 2026](#)). The Finnish null ([Kauhanen and Rouvinen, 2026](#)) is attributed to demographics.

When formal methods control for firm-level confounders, the magnitudes tend to be smaller. [Brynjolfsson et al. \(2025\)](#) find −16 log points in a Poisson event study with firm fixed effects, compared with larger raw occupation-level declines. [Lodefalk et al. \(2026\)](#) find −5.5% within employer by 2025H1, and their job-posting analysis suggests that much of the aggregate decline in AI-exposed occupations reflects the Riksbank rate cycle. [Humlum and Vestergaard \(2026\)](#) document an aggregate Danish decline in early-career employment that resembles the US and Norwegian patterns, but their firm-level difference-in-differences, the only study with individual-level AI adoption data, does not find that workplace AI chatbot adoption drives the decline, bounding the earnings effect at less than 2 to 3 percent.

The gap between descriptive and formal estimates need not imply that AI plays no role. Formal methods absorb variation that may include both confounders and genuine AI effects operating through channels other than direct firm-level adoption (for example, competitive pressure from AI-adopting rivals, or supply-side reallocation by workers anticipating AI disruption). Nonetheless, the pattern suggests that aggregate descriptive trends, including those we report for Norway, should be interpreted with caution. Part of the observed decline may reflect concurrent forces such as monetary tightening, post-COVID corrections, or seniority dynamics rather than AI-specific adjustment. At the industry level, [Bick et al. \(2026\)](#) find no clear evidence that AI adoption is associated with employment changes in either Europe or the US, consistent with the wage and employment nulls in [Humlum and Vestergaard \(2026\)](#). Norway, Sweden, Denmark, and Finland share similar labor market institutions, yet the formal results differ across countries, so institutional structure alone does not explain the heterogeneity.

Table 6: Cross-Country Comparison of Early Labor Market Patterns under Generative AI

Country	Study	Method	Direction (22–25)	Notes
<i>Panel A: Descriptive time series</i>				
US	Brynjolfsson et al. (2025)	Descriptive	Decline	SW developers 22–25: –20%; top-2 quintiles: –6%
Sweden	Lodefalk et al. (2026)	Descriptive	Decline	≈ –44% raw in top quartile (Fig. A23)
Denmark	Humlum & Vestergaard (2026)	Descriptive	Decline	Aggregate early-career decline mirrors US (App. D.2)
Finland	Kauhanen & Rouvinen (2026)	Descriptive	Null	Trends reflect demographics, not AI
UK	Teeselink (2025)	Descriptive	Decline	High-wage, junior positions
<i>Panel B: Formal estimation</i>				
US	Brynjolfsson et al. (2025)	Poisson event study, firm FE	Decline	–16 log pts, most vs. least exposed quintile
Sweden	Lodefalk et al. (2026)	DiD, employer FE	Decline	–5.5% within-employer by 2025H1; posting decline driven by rate hikes, not AI
Denmark	Humlum & Vestergaard (2026)	DiD, worker/workplace FE	Null	Aggregate decline not driven by firm AI adoption; rules out > 2% earnings effect
UK	Teeselink (2025)	DiD	Decline	Junior positions at high-wage firms

Notes: Studies sorted by year within each panel. “Direction” refers to the qualitative sign of the relative employment change for workers aged 22–25 in the most AI-exposed occupations after November 2022. Panel A reports descriptive patterns (raw or per-capita trends); Panel B reports results from formal econometric methods with controls and fixed effects. The descriptive-formal gap is substantial in all studies that employ both: Lodefalk et al. find –44% raw vs. –5.5% in the employer-FE event study; Humlum & Vestergaard find an aggregate decline descriptively but a precise null when using firm-level adoption as treatment. Magnitudes across rows use different units and baselines and are not directly comparable.

6.2 Alternative Explanations

Four non-AI channels could produce the same age-by-exposure gradient, and the aggregate data cannot fully separate them.

Post-COVID technology correction. Software, data analysis, and digital services hired aggressively from 2020 through 2022, disproportionately absorbing young workers. The subsequent correction, as pandemic-era demand for digital services normalized, would reproduce the post-2022 pattern with no AI involvement. The 22–25 software developer line in Section 4 starts to slope down before ChatGPT’s release, consistent with this channel.

Monetary tightening. The Norges Bank rate cycle (Section 3.2) overlaps the post-ChatGPT window. Higher rates depress labor demand in finance, real estate, construction, and technology, sectors that also overlap substantially with the high-exposure quintiles, and that shed junior workers first. The public-sector null fits a market-driven story but cannot separate AI from rates at the aggregate level.

Seniority dynamics. In any downturn, employers in technology-intensive sectors retain experienced workers and cut entry-level hiring, producing a mechanical age gradient. Separating this from AI-driven substitution requires within-firm variation, which is exactly what the firm-FE design in Section 4.6 provides. There, the within-firm Q5-vs-Q1 coefficient for 22–25 is near zero, so the cohort-level Q5 decline runs through which firms are hiring rather than which workers existing firms retain. That pattern fits hiring freezes better than internal substitution.

Supply-side reallocation. Workers entering the labor market after 2022 may be avoiding occupations they perceive as exposed to AI. This would generate the same young-worker pattern in exposed occupations through worker choice rather than employer substitution.

The Q5–Q1 gradient is a relative pattern within the 22–25 age band; the overall 22–25 line in Figure 2 is roughly flat, so Q5 losses are largely offset by Q1–Q4 stability. The aggregate data alone cannot distinguish occupational reallocation among young workers from outright exit. The four channels above and the AI channel can all be operating at once.

6.3 Implications

The public-sector null and the wage null are consistent with Norway’s two-tier bargaining system ([Bhuller et al., 2022](#)) channelling any AI-related adjustment toward the hiring margin in the private sector. Downward-rigid wage floors limit short-run wage adjustment; the locally negotiated drift channel could deliver a wage response over a longer horizon. The qualitative pattern holds across all four exposure measures despite their different conceptual foundations, while magnitudes vary with measure choice; the field has not converged on a standard “AI exposure” definition, and the results should be read with that measurement uncertainty in mind.

7 Conclusion

This paper documents a post-2022 age-exposure gradient in Norwegian employment using aggregated cell counts from population administrative data (2021–2026). The gradient is concentrated in the private sector, has no corresponding wage divergence, and is present across four distinct AI exposure measures.

AI-related adjustment, the post-COVID correction in technology hiring, monetary tightening by Norges Bank, and seniority-based retention could each generate the same age-exposure gradient, and the aggregate data we have cannot tell them apart. [Humlum and Vestergaard \(2026\)](#) document the same aggregate pattern in Denmark and show that observable workplace adoption of AI chatbots accounts for at most a small fraction of it. Worker- and firm-level designs in [Brynjolfsson et al. \(2025\)](#), [Lodefalk et al. \(2026\)](#), and [Humlum and Vestergaard \(2026\)](#) answer related questions at finer units of observation.

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Appendix A: Additional Figures

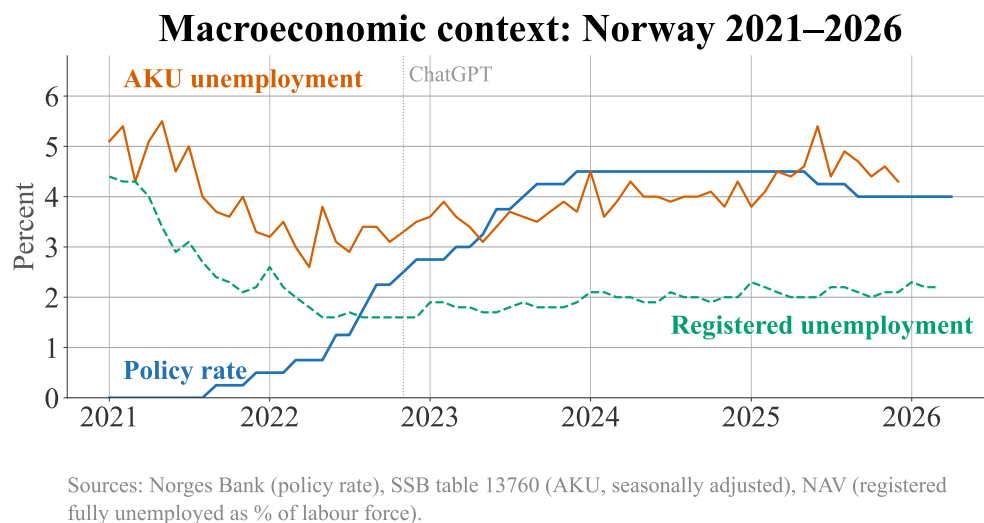


Figure A1: Macroeconomic context: Norges Bank policy rate and registered unemployment rate, 2015–2025.

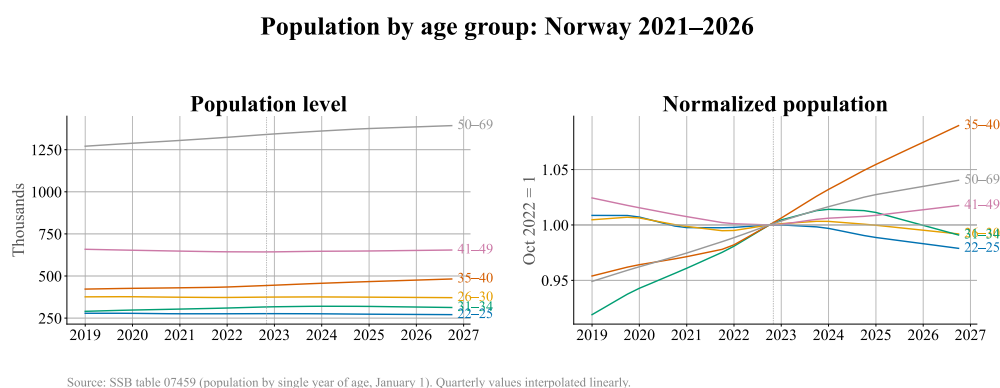
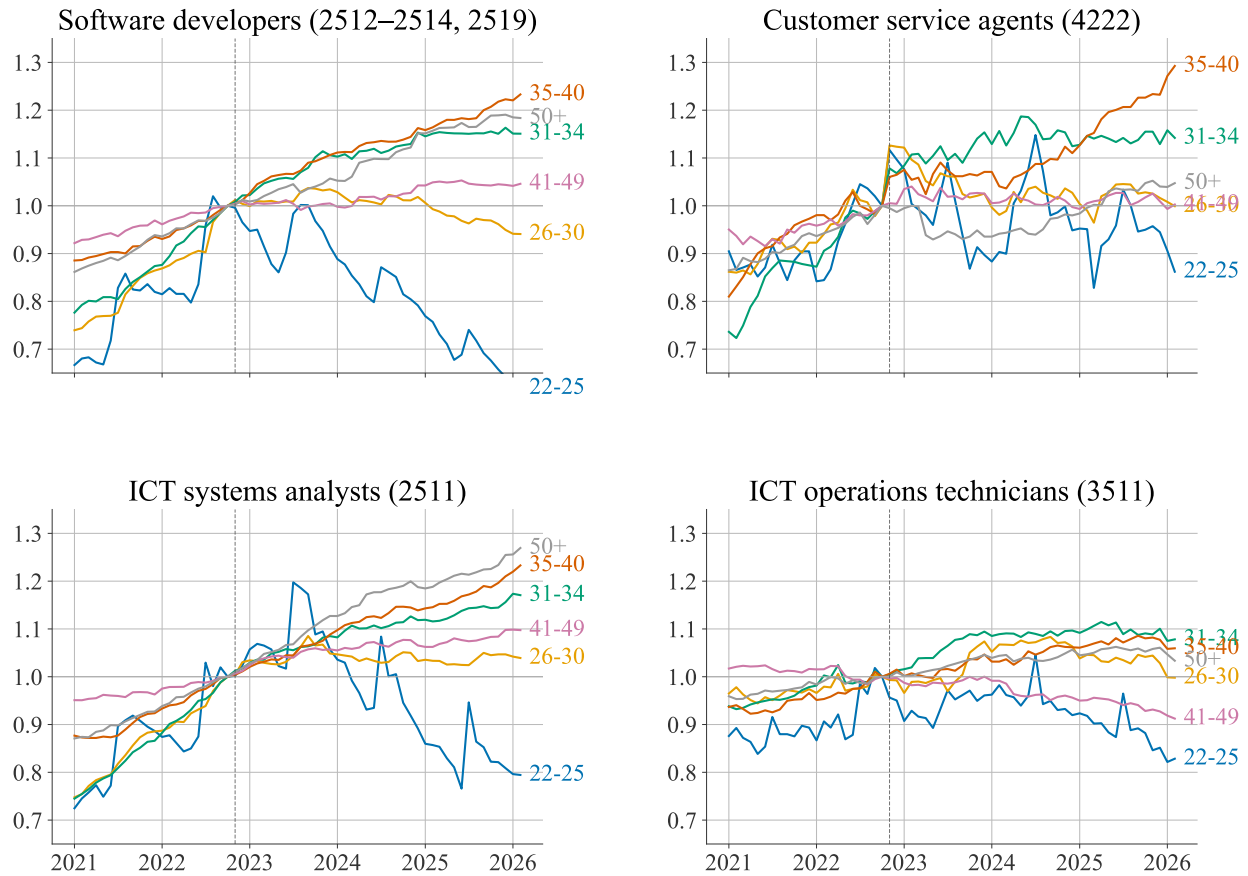


Figure A2: Resident population by age group, 2015–2025.

Appendix B: Job Dimensionality Split

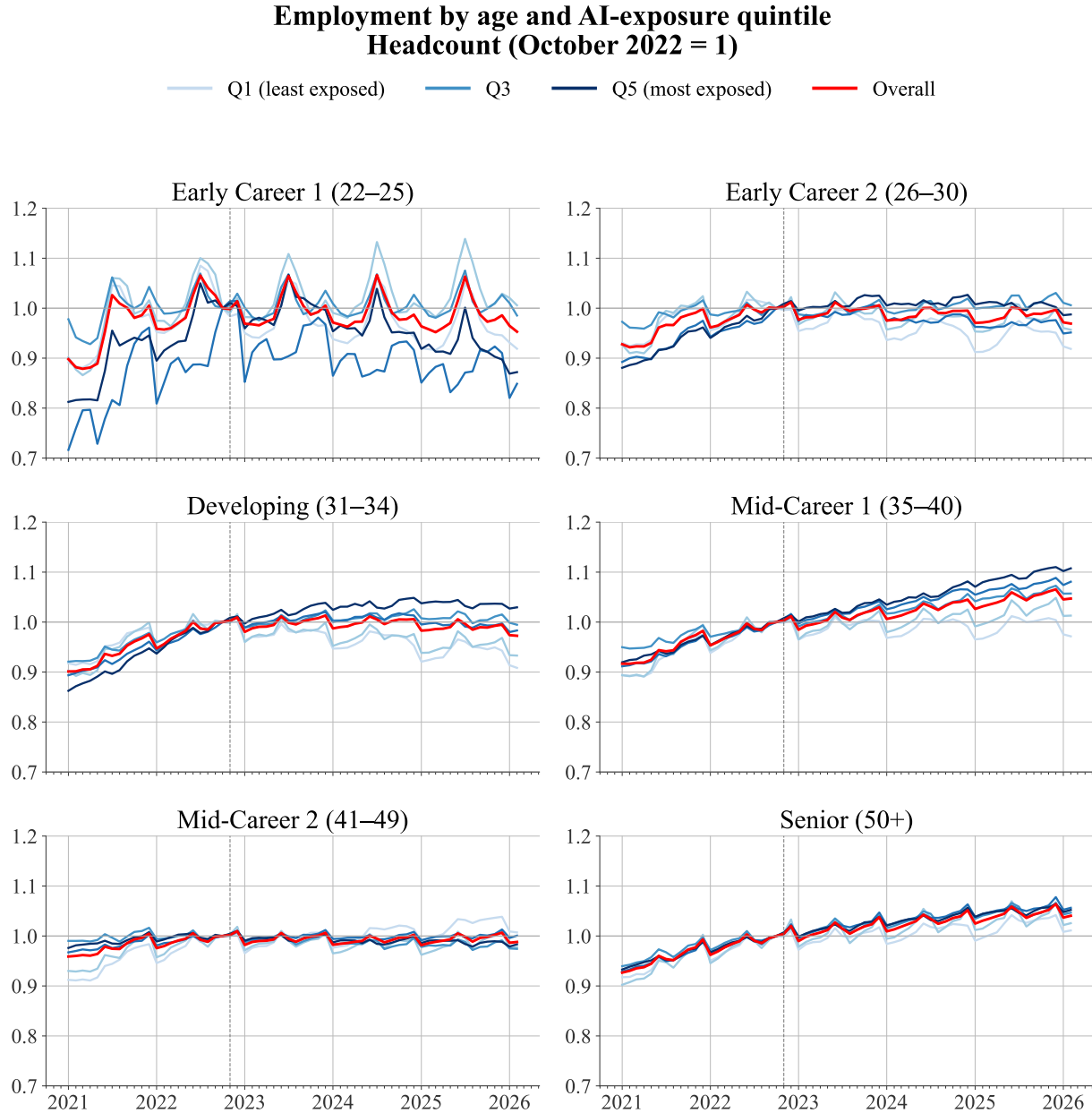
Figure A22 addresses the O-ring / job-dimensionality prediction of Gans and Goldfarb (2025) and Imas and Shukla (2026): when tasks are quality complements rather than independent, average exposure is a poor summary of adjustment, and any movement should concentrate in occupations with high exposure and a small task set, where automation of a few tasks covers most of the job. We split occupations at the median on Eloundou et

Employment by age group Headcount (October 2022 = 1)



Notes: Employment counts from Norwegian A-ordningen register data, normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). STYRK-08 occupation codes in parentheses. Age is calculated using month of birth.

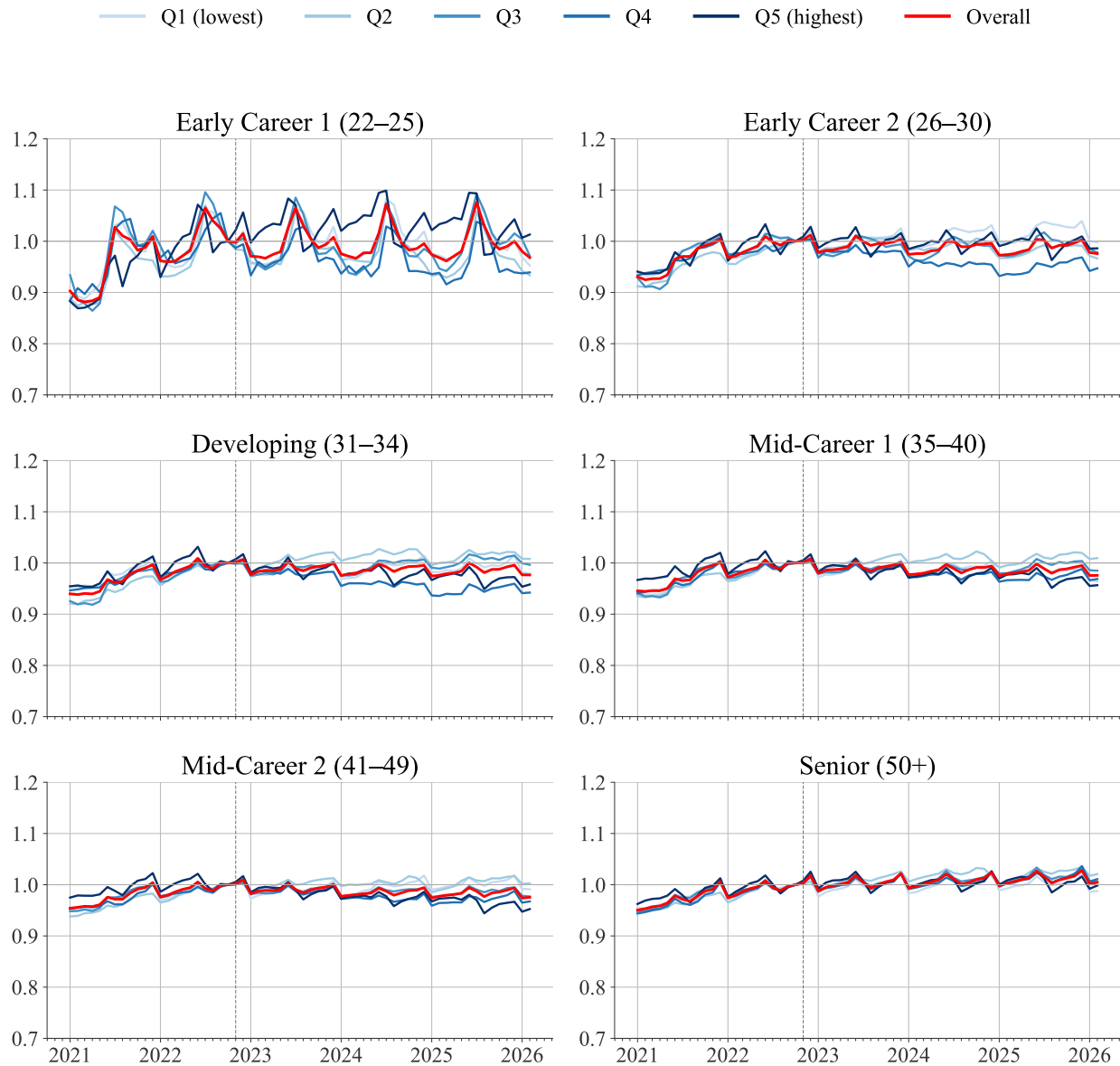
Figure A3: Unadjusted employment (raw headcount) in selected occupations by age group, October 2022 = 1. Compare with the population-adjusted Figure 1 used in the main text.



Notes: Employment counts from Norwegian A-ordningen register data, normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Exposure quintiles based on GPT-4 β measures from Eloundou et al. (2024), mapped to STYRK-08 via SOC 2010/ISCO-08 crosswalk (BLS, 2012). STYRK-08 = ISCO-08 at 4-digit level (SSB Notater 17/2011). Darker lines indicate higher AI exposure.

Figure A4: Unadjusted employment (raw headcount) by Eloundou et al. exposure quintile and age group, October 2022 = 1. Compare with the population-adjusted Figure 2 used in the main text.

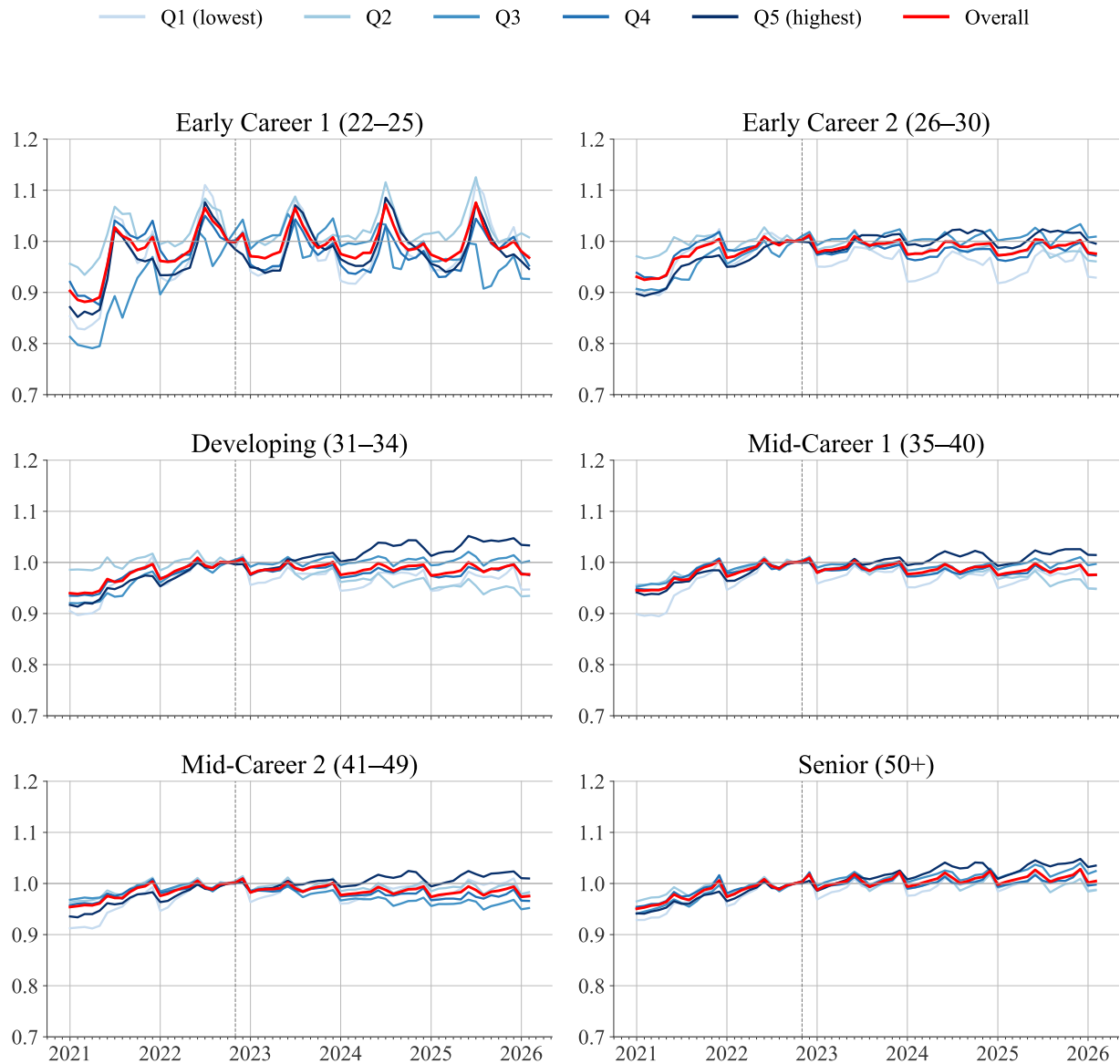
Employment per capita by age and Handa et al. automation quintile (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1), normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Quintiles based on Handa et al. automation, from the Anthropic Economic Index, mapped to STYRK-08 via SOC 2010/ISCO-08 crosswalk (BLS, 2012). STYRK-08 = ISCO-08 at 4-digit level (SSB Notater 17/2011). Darker lines indicate higher values.

Figure A5: Employment by age and Handa et al. automation exposure quintile, October 2022 = 1.

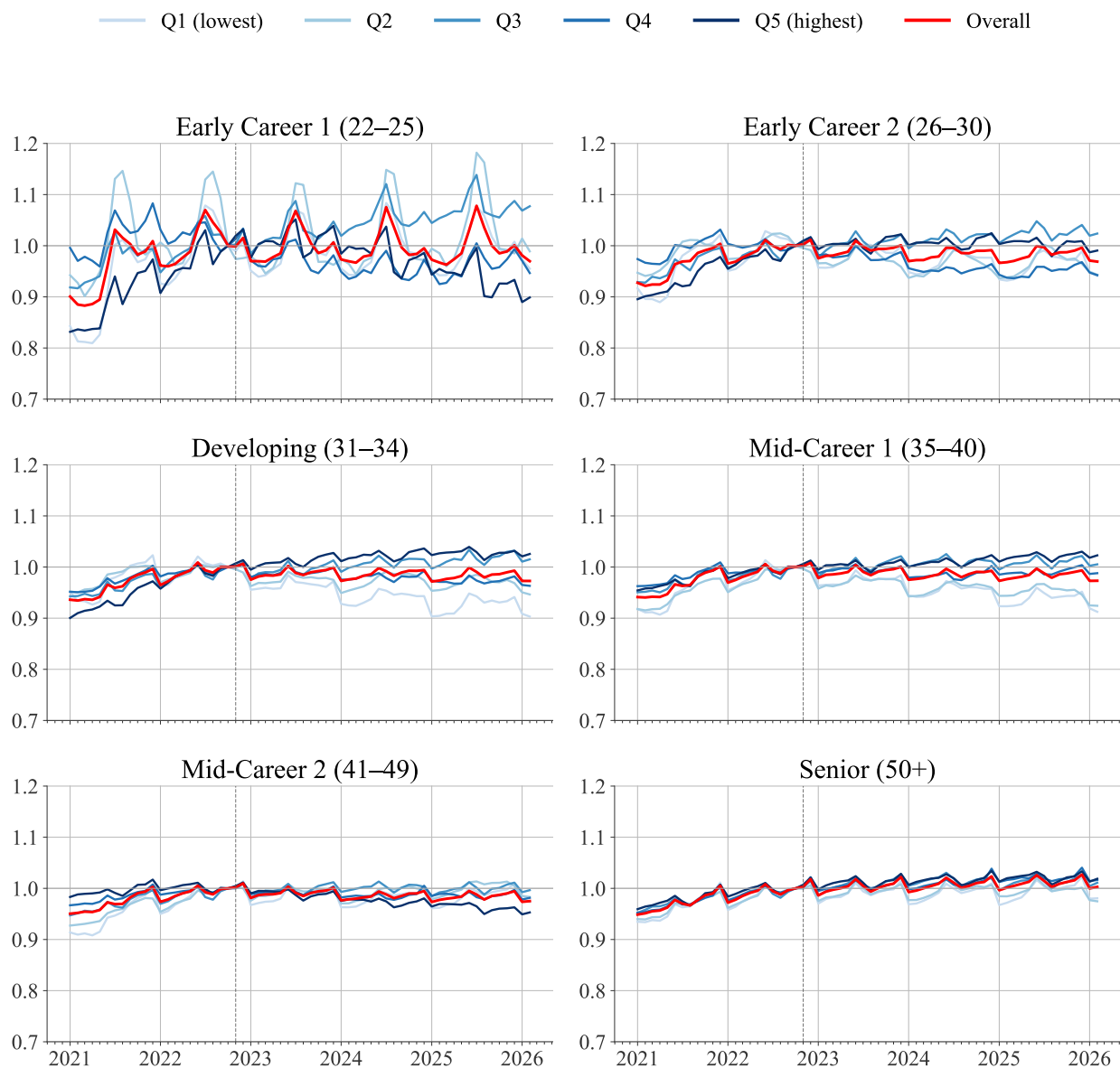
Employment per capita by age and Handa et al. augmentation quintile (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1), normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Quintiles based on Handa et al. augmentation, from the Anthropic Economic Index, mapped to STYRK-08 via SOC 2010/ISCO-08 crosswalk (BLS, 2012). STYRK-08 = ISCO-08 at 4-digit level (SSB Notater 17/2011). Darker lines indicate higher values.

Figure A6: Employment by age and Handa et al. augmentation exposure quintile, October 2022 = 1.

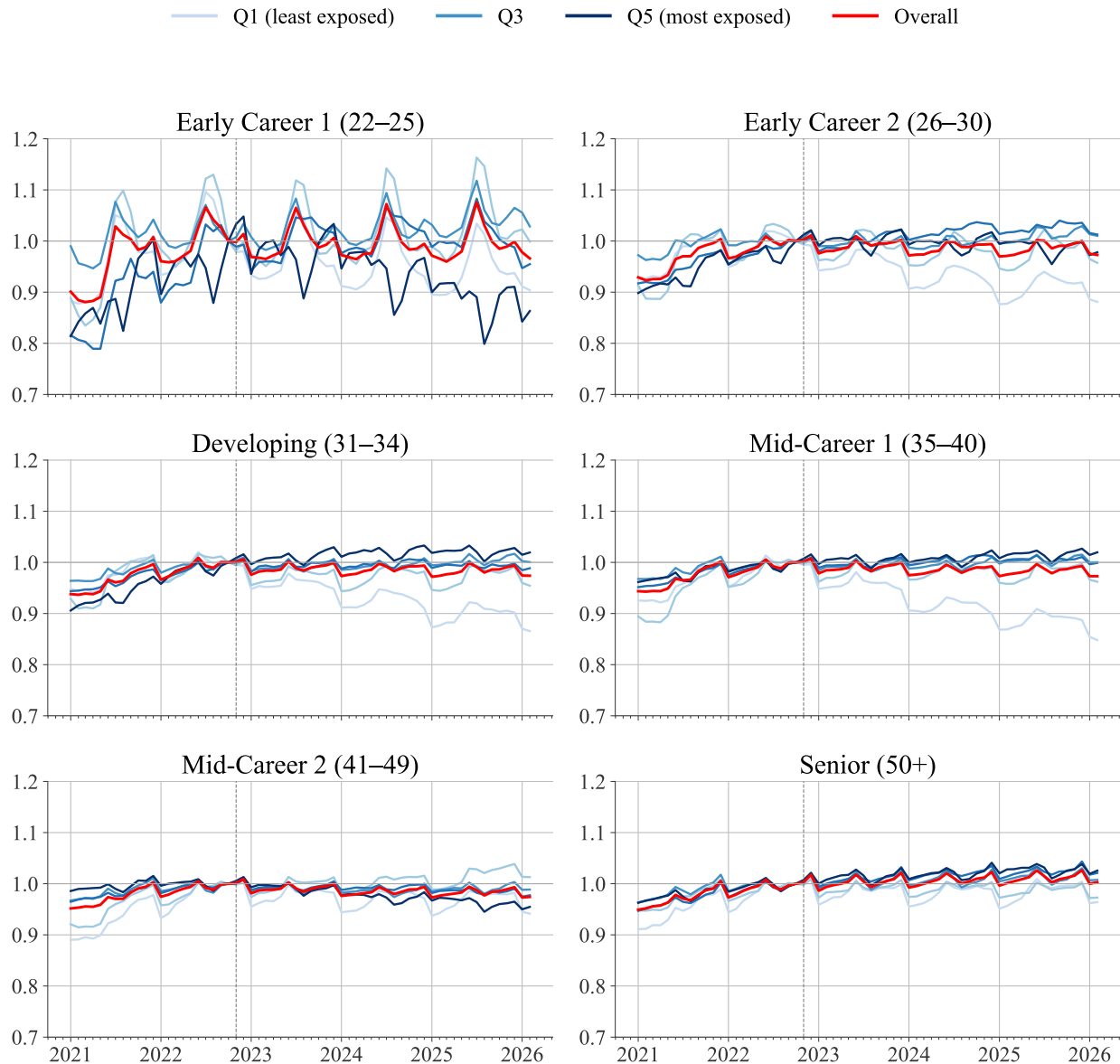
Employment per capita by age and Anthropic job exposure (time-weighted) quintile (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1), normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Quintiles based on Anthropic job exposure (time-weighted), from the Anthropic Economic Index, mapped to STYRK-08 via SOC 2010/ISCO-08 crosswalk (BLS, 2012). STYRK-08 = ISCO-08 at 4-digit level (SSB Notater 17/2011). Darker lines indicate higher values.

Figure A7: Employment by age and Anthropic (2026) job_exposure quintile, October 2022 = 1. The job_exposure measure is a time-weighted version of Claude usage with an automation penalty, released by Anthropic in March 2026.

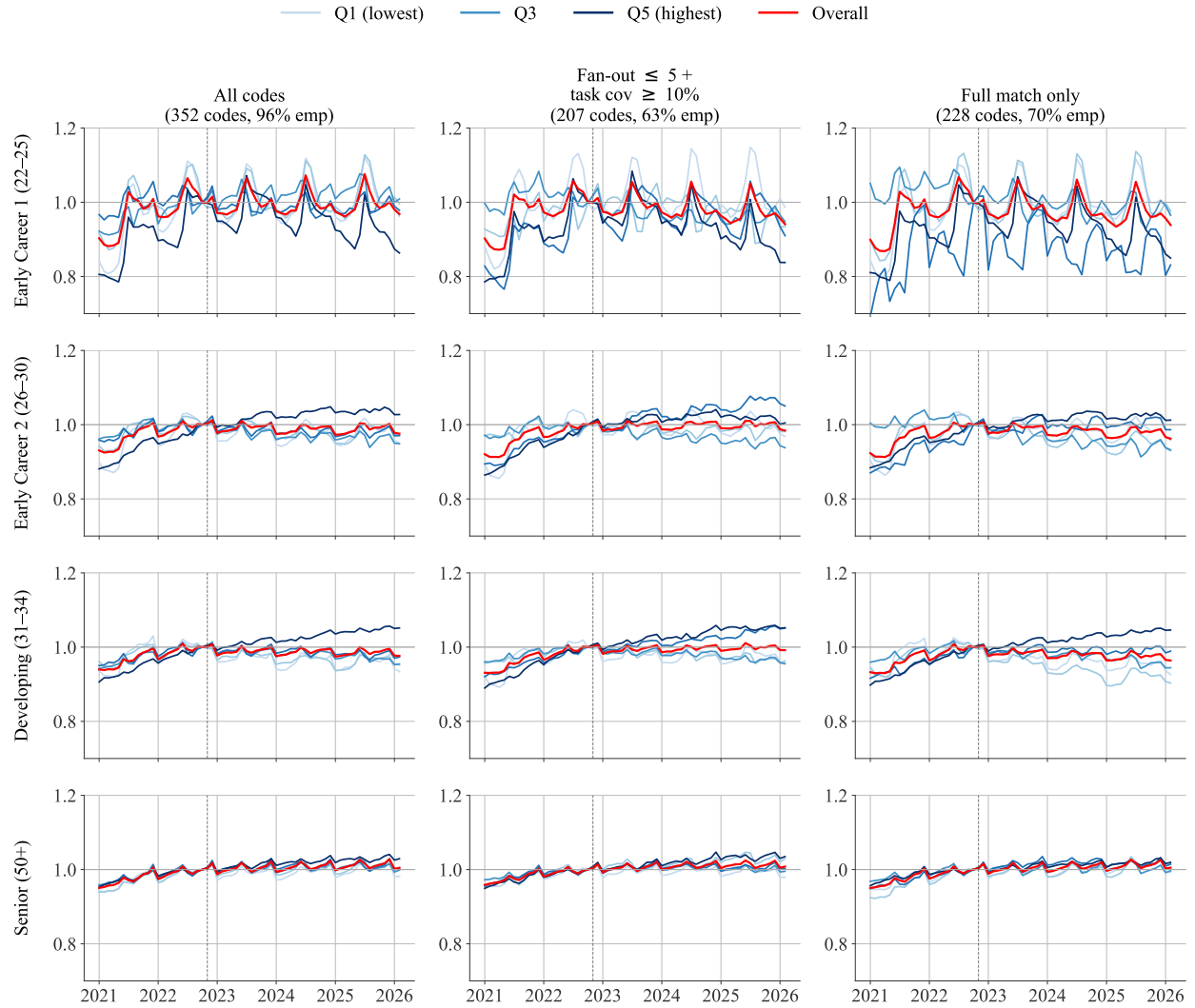
Employment per capita by age and Felten et al. AIOE Language Modeling quintile (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1), normalized to October 2022 = 1. Felten et al. AIOE Language Modeling quintiles from Felten et al. (2021), mapped to STYRK-08 via SOC 2010/ISCO-08 crosswalk. Darker lines indicate higher AI exposure.

Figure A8: Employment by age and Felten AIOE Language Modeling quintile, October 2022 = 1.

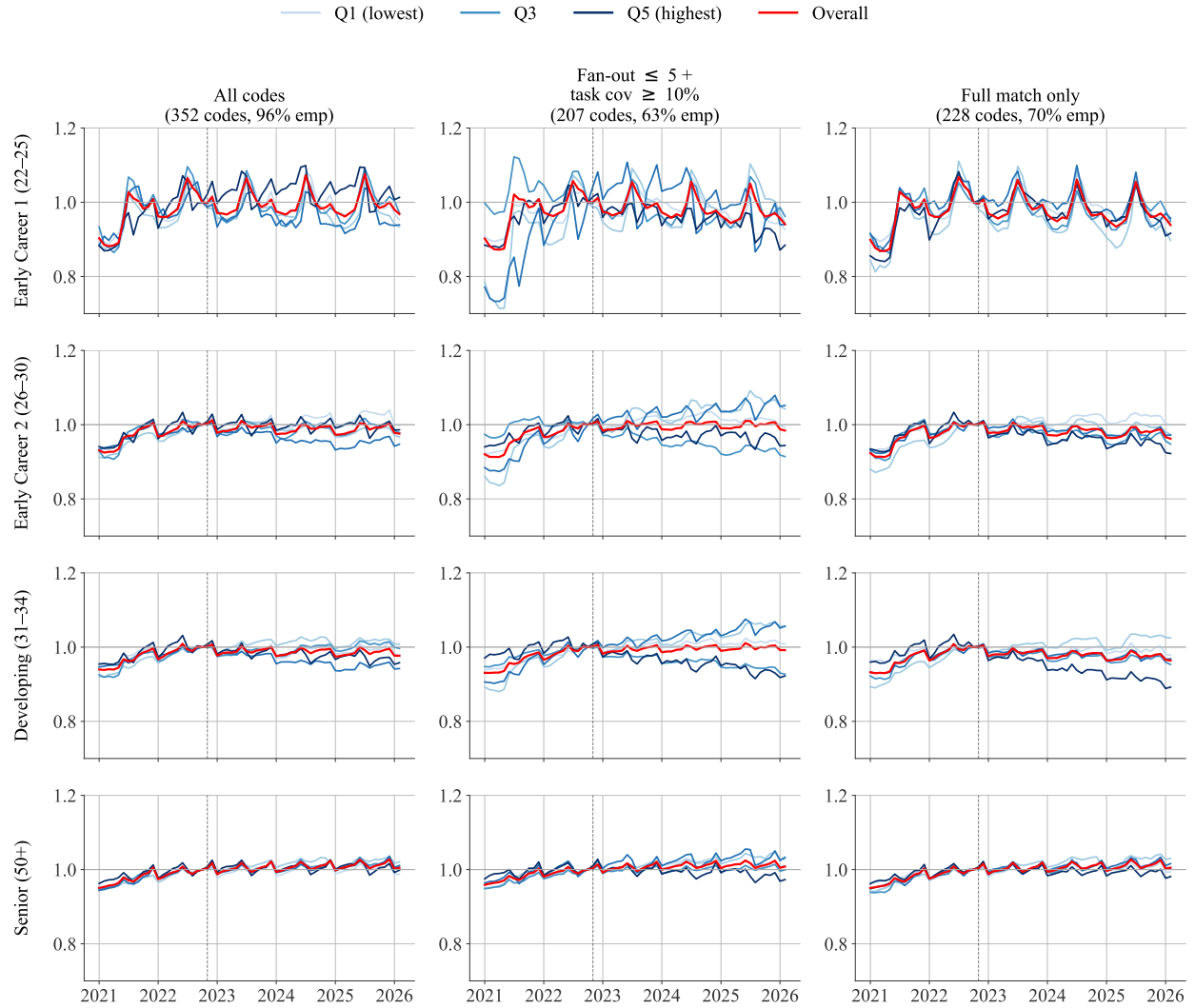
Sensitivity: Handa et al. overall exposure quintiles by crosswalk quality



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1). Sensitivity of Figure 5a to crosswalk quality restrictions. Left: all mapped codes. Center: fan-out ≤ 5 and task coverage $\geq 10\%$. Right: only full SOC-ISCO matches (fan-out ≤ 2). Quintiles re-assigned within each restricted sample. Handa et al. (2025) Anthropic Economic Index overall exposure.

Figure A9: Crosswalk quality sensitivity for Handa et al. overall exposure quintiles.

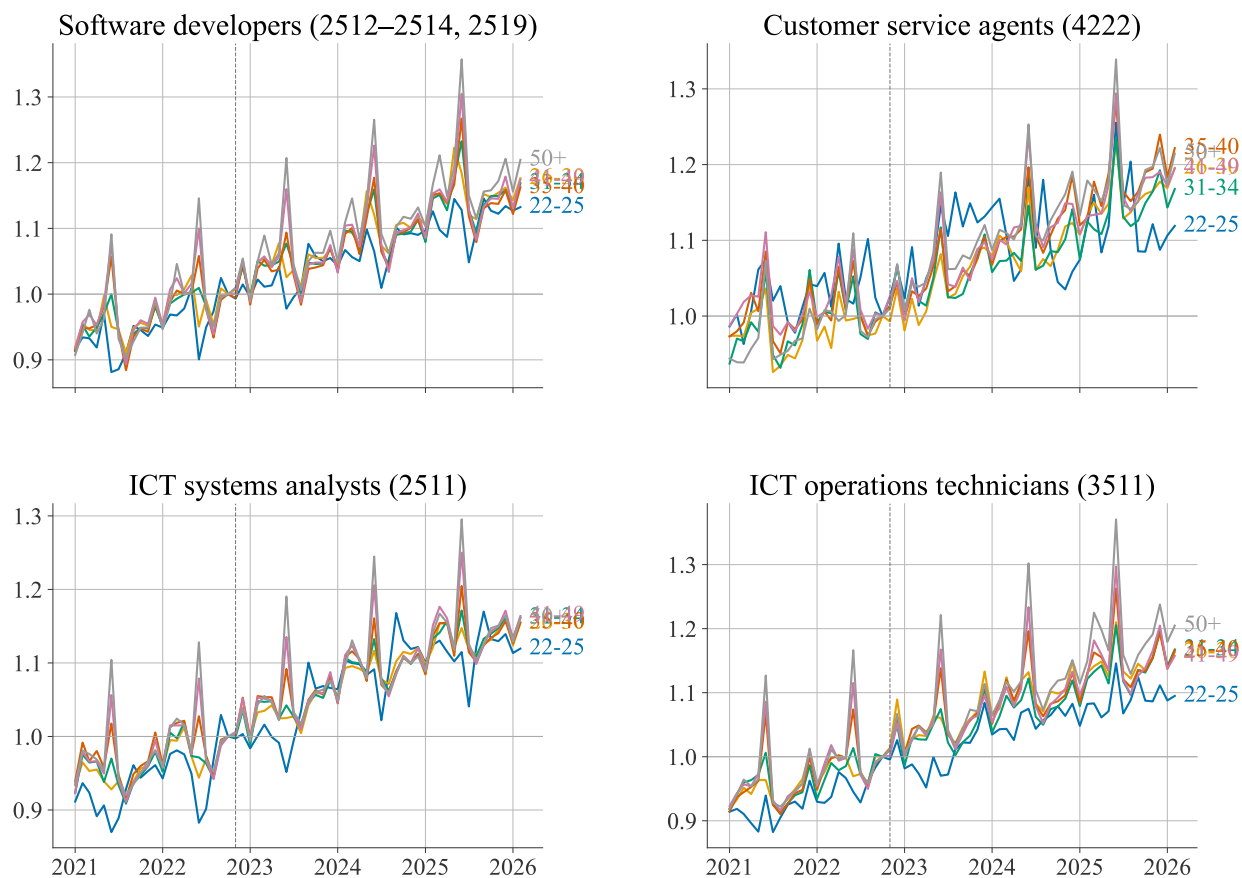
Sensitivity: Handa et al. automation quintiles by crosswalk quality



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1). Sensitivity of Figure 5b to crosswalk quality restrictions. Left: all mapped codes. Center: fan-out ≤ 5 and task coverage $\geq 10\%$. Right: only full SOC-ISCO matches. Quintiles re-assigned within each restricted sample. Handa et al. (2025) Anthropomic Economic Index automation share.

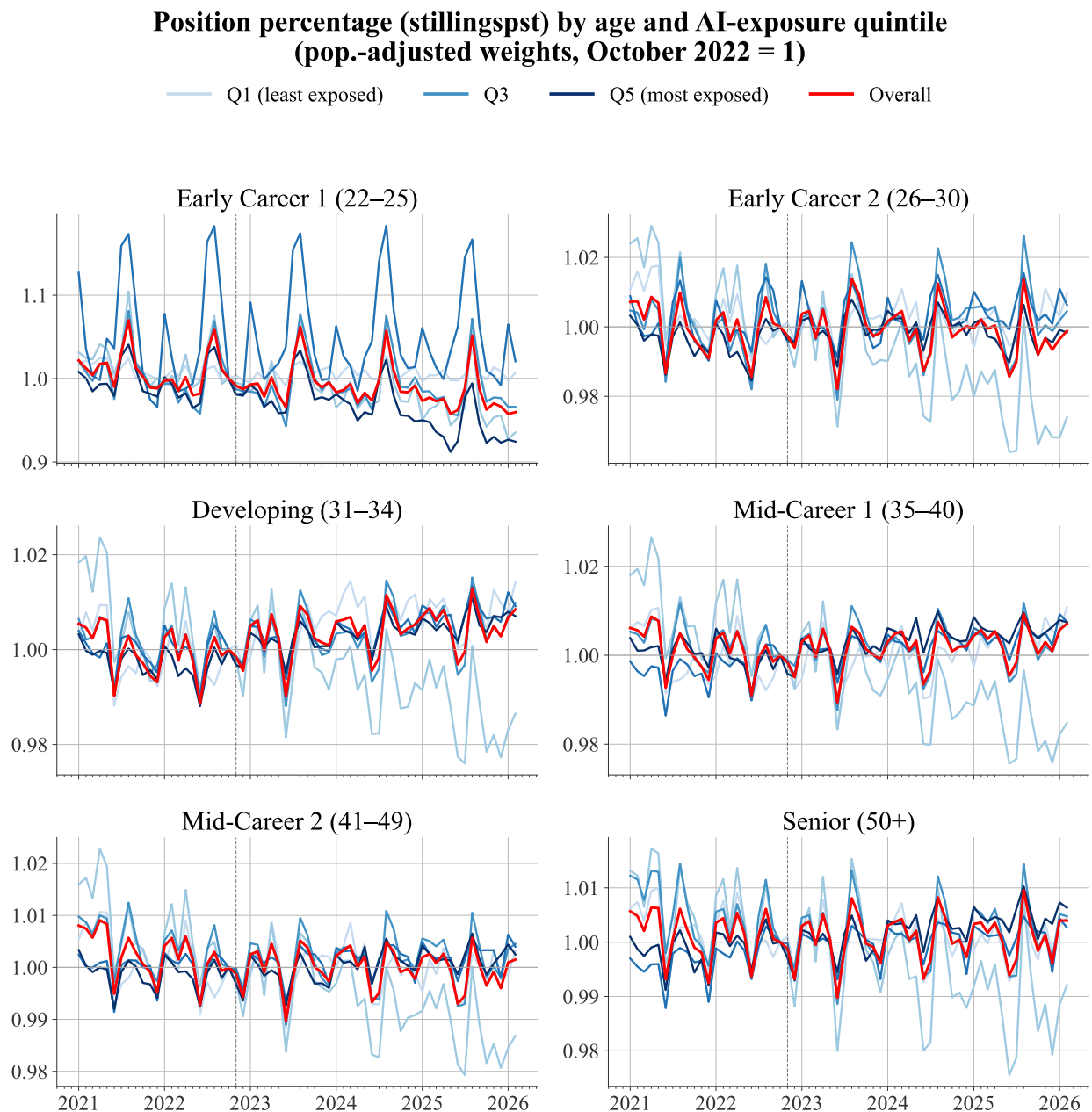
Figure A10: Crosswalk quality sensitivity for Handa et al. automation exposure quintiles.

**Monthly earnings (kontantlønn) by age group (pop.-adjusted weights)
(October 2022 = 1)**



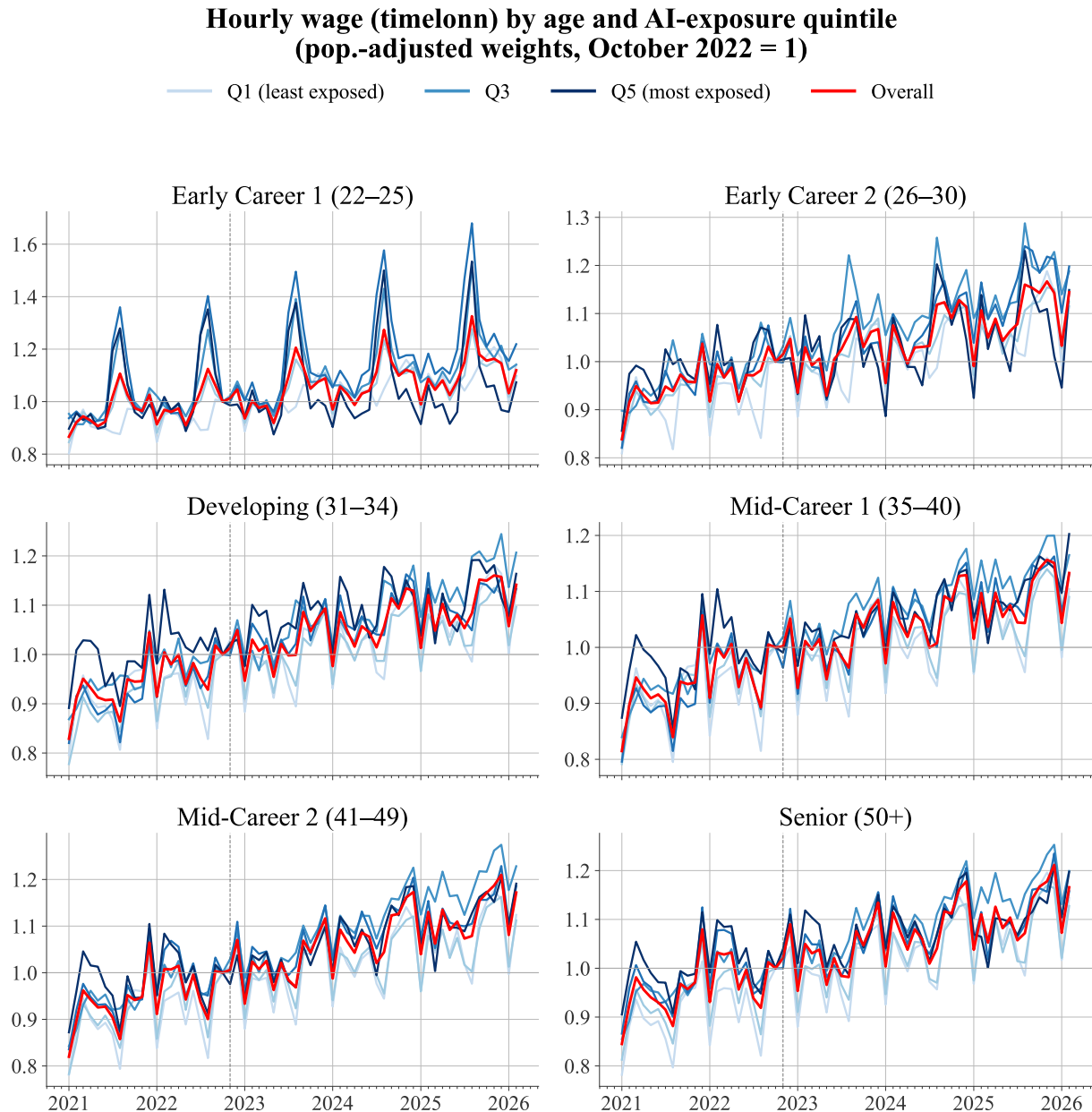
Notes: Mean kontantlønn from Norwegian A-ordningen register data, normalized to October 2022 = 1.
Weights are employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1).
Dashed vertical line marks the launch of ChatGPT (November 2022). Multi-code occupations are employment-weighted.

Figure A11: Monthly earnings by age group for selected occupations. October 2022 = 1.



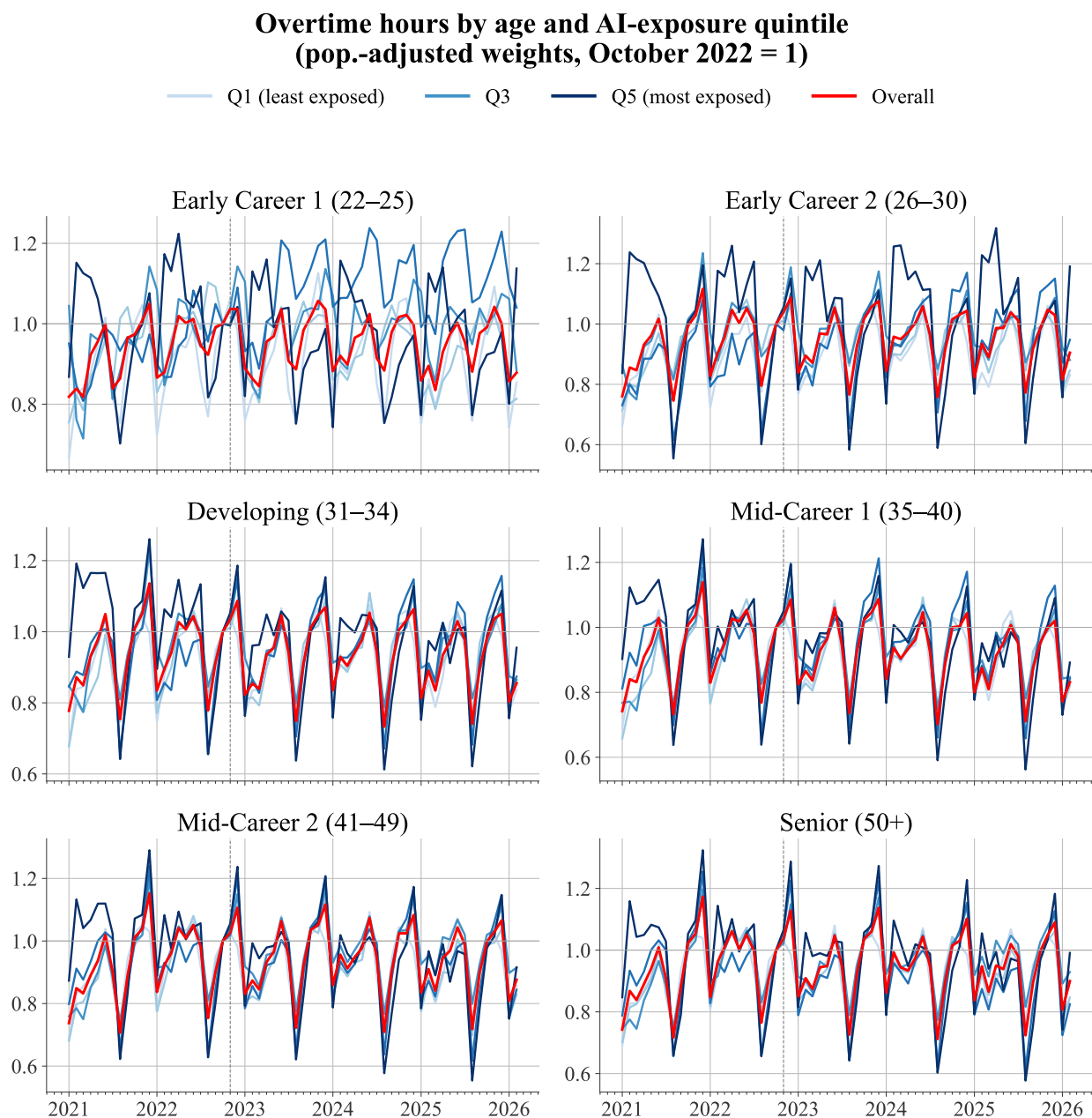
Notes: Employment-weighted mean stillingspst from Norwegian A-ordningen register data, normalized to October 2022 = 1. Weights are employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1). Dashed vertical line marks the launch of ChatGPT (November 2022). Exposure quintiles based on Eloundou et al. (2024) GPT-4 beta.

Figure A12: Position percentage by Eloundou et al. quintile and age group. October 2022 = 1.



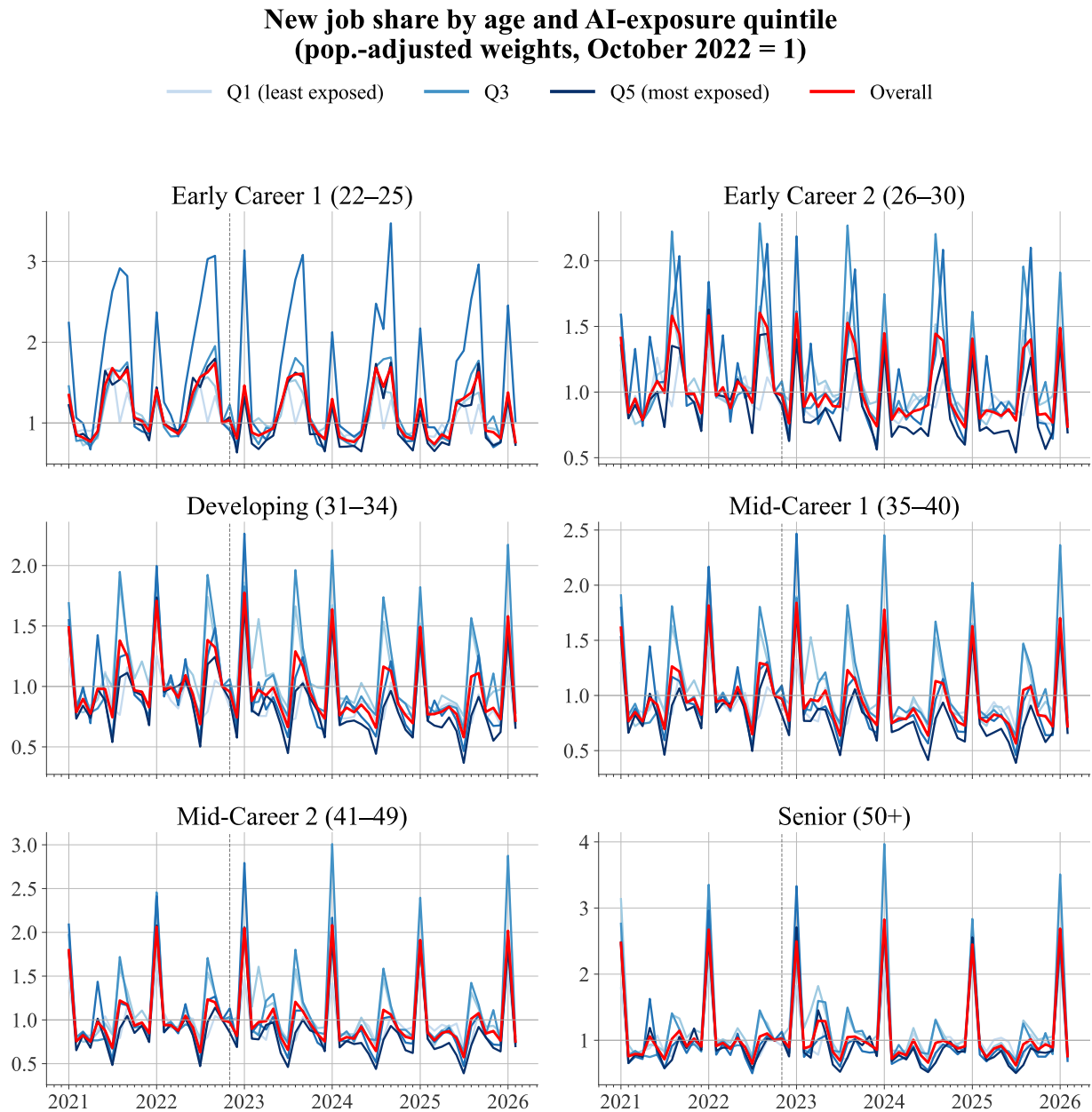
Notes: Employment-weighted mean timelonn from Norwegian A-ordningen register data, normalized to October 2022 = 1. Weights are employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1). Dashed vertical line marks the launch of ChatGPT (November 2022). Exposure quintiles based on Eloundou et al. (2024) GPT-4 beta.

Figure A13: Hourly wage by Eloundou et al. quintile and age group. October 2022 = 1.



Notes: Employment-weighted mean `overtid_timer` from Norwegian A-ordningen register data, normalized to October 2022 = 1. Weights are employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1). Dashed vertical line marks the launch of ChatGPT (November 2022). Exposure quintiles based on Eloundou et al. (2024) GPT-4 beta.

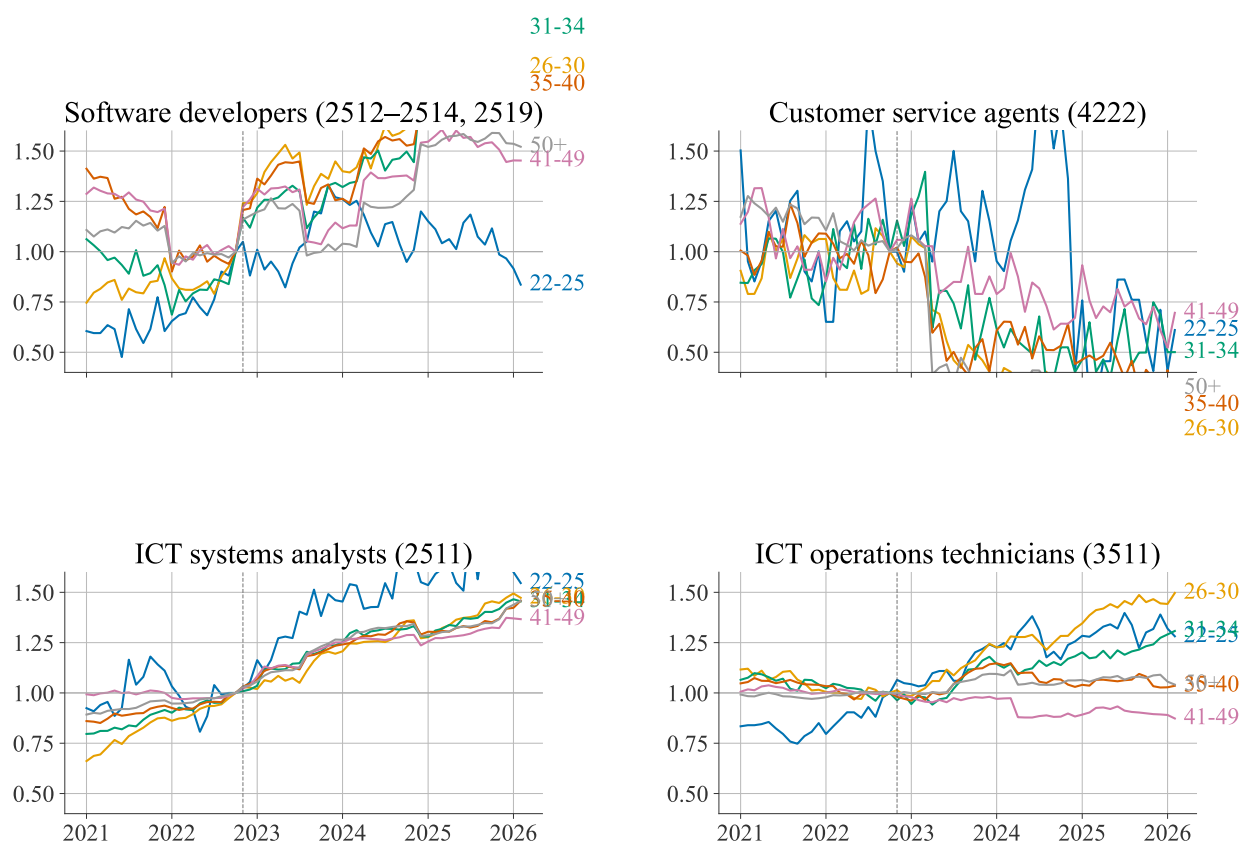
Figure A14: Overtime hours by Eloundou et al. quintile and age group. October 2022 = 1.



Notes: Employment-weighted mean `ny_jobb` from Norwegian A-ordningen register data, normalized to October 2022 = 1. Weights are employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1). Dashed vertical line marks the launch of ChatGPT (November 2022). Exposure quintiles based on Eloundou et al. (2024) GPT-4 beta.

Figure A15: New job share by Eloundou et al. quintile and age group. October 2022 = 1.

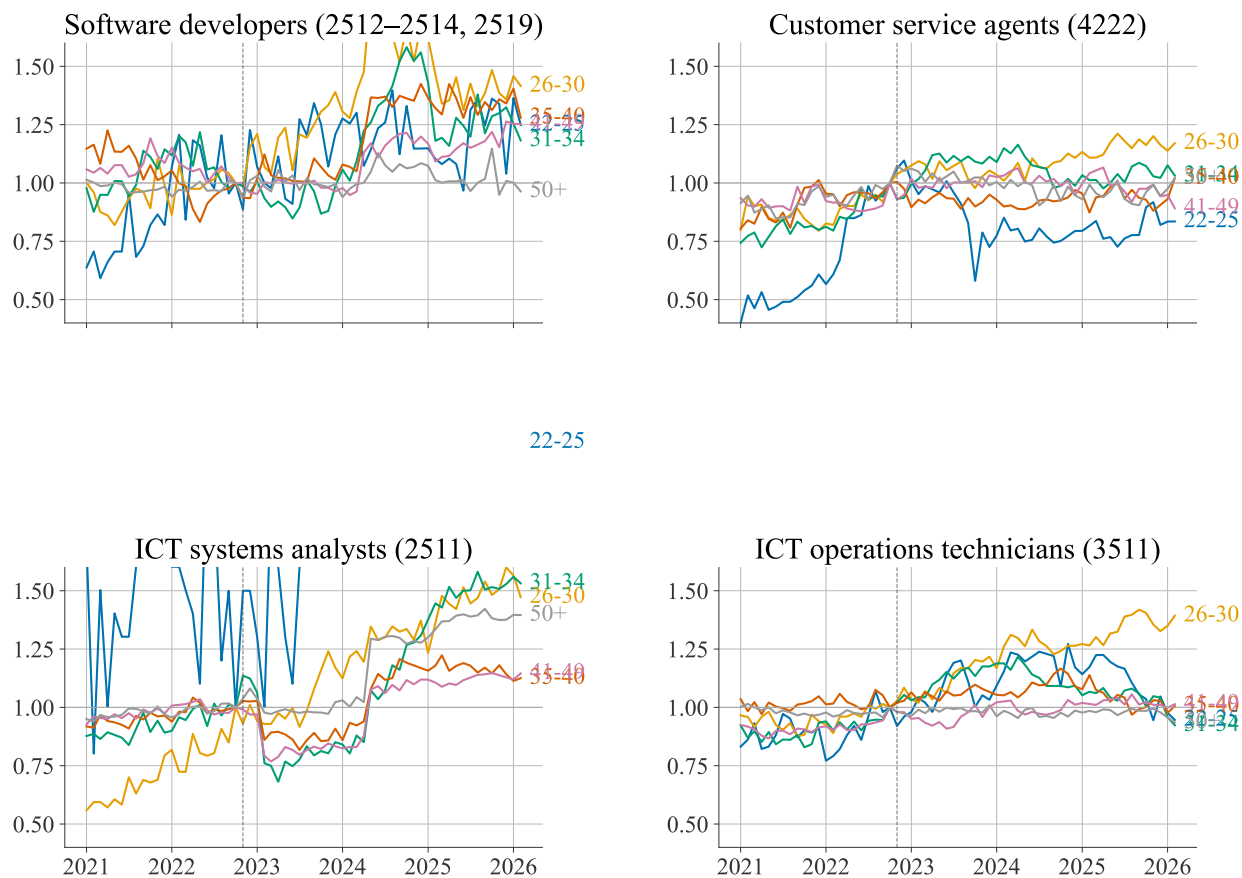
Employment per capita by age group, State sector (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1) in state sector, normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Sector: State (1110,1120,6100), Municipal (1510,1520,6500), Private (other).

Figure A16: Employment by age group in selected occupations, state sector. October 2022 = 1.

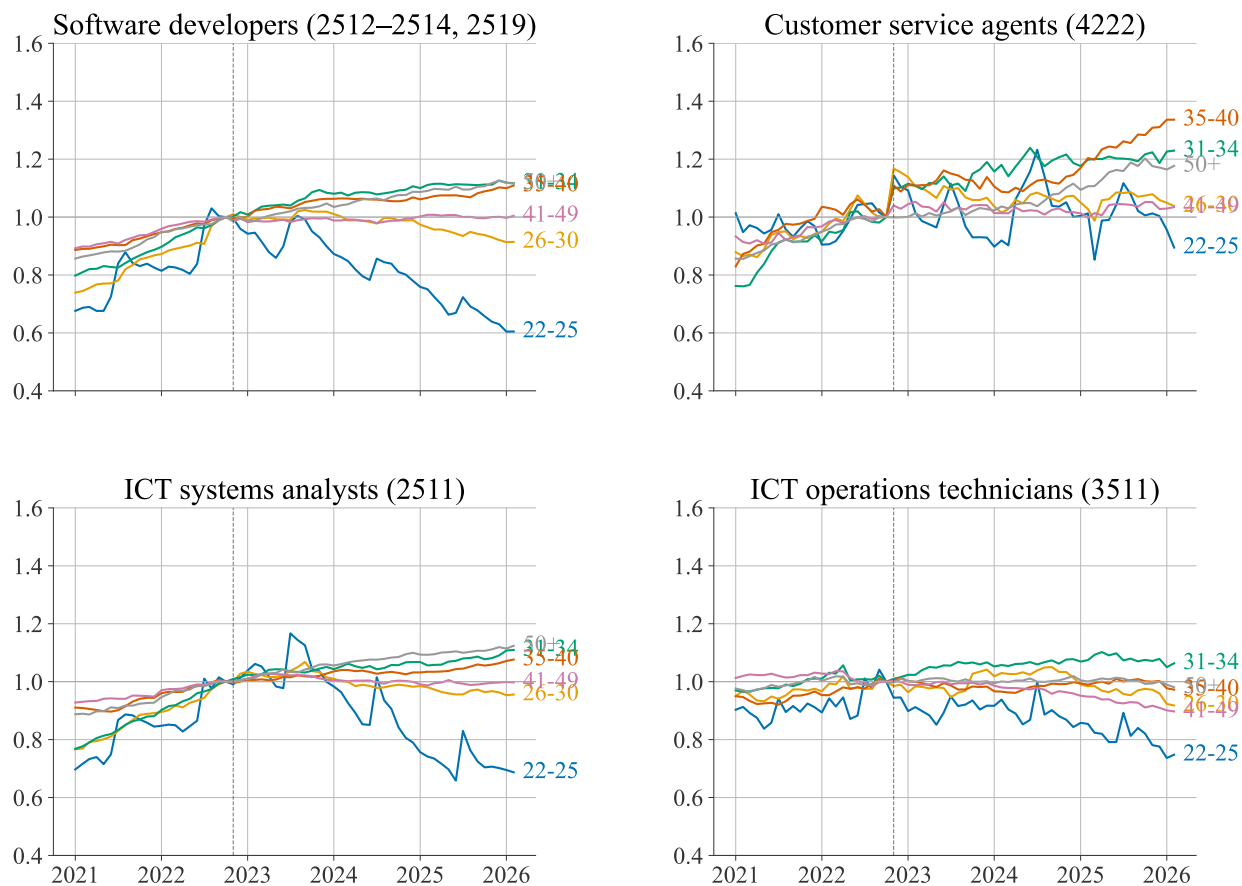
Employment per capita by age group, Municipal sector (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1) in municipal sector, normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Sector: State (1110,1120,6100), Municipal (1510,1520,6500), Private (other).

Figure A17: Employment by age group in selected occupations, municipal sector. October 2022 = 1.

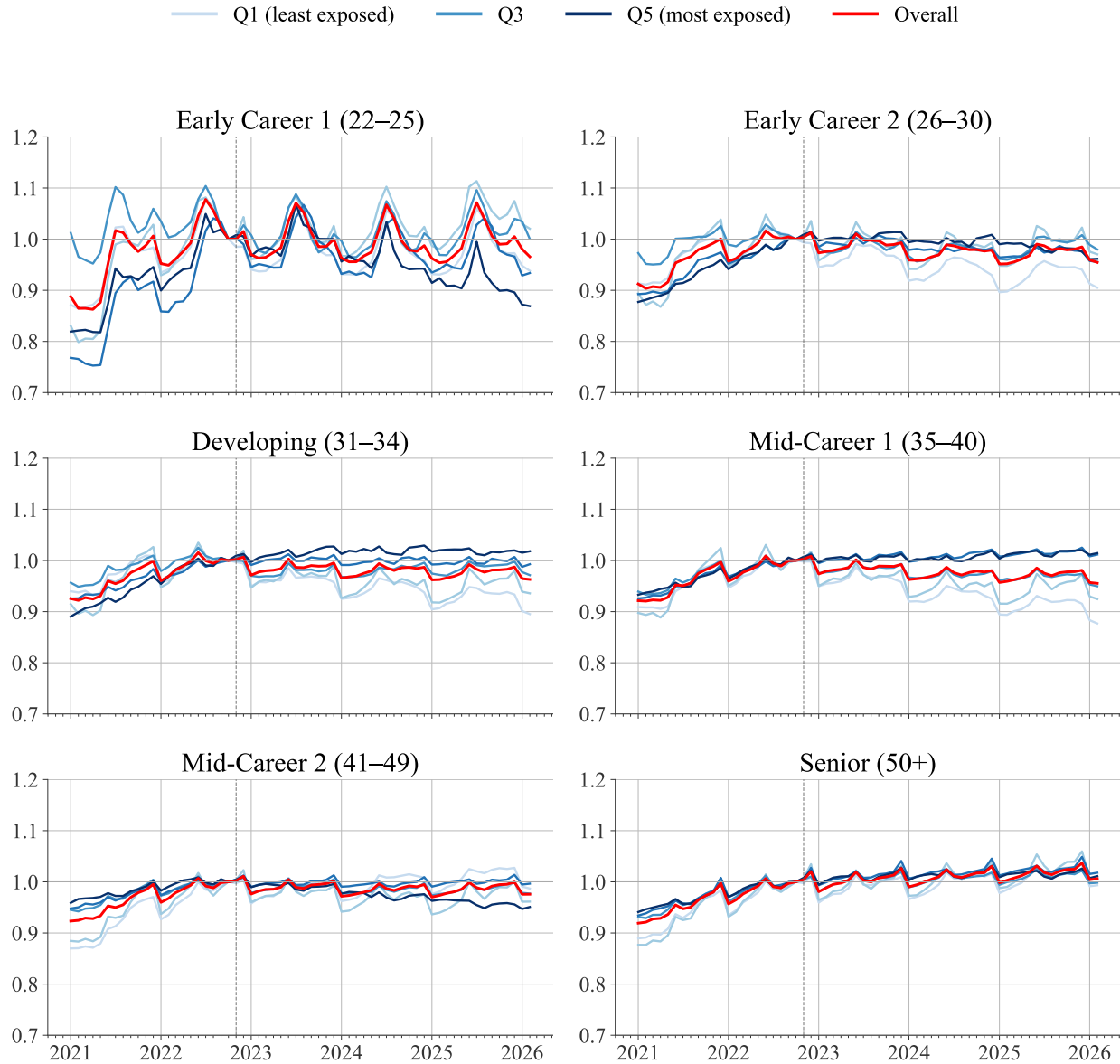
Employment per capita by age group, Private sector (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1) in private sector, normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Sector: State (1110,1120,6100), Municipal (1510,1520,6500), Private (other).

Figure A18: Employment by age group in selected occupations, private sector. October 2022 = 1.

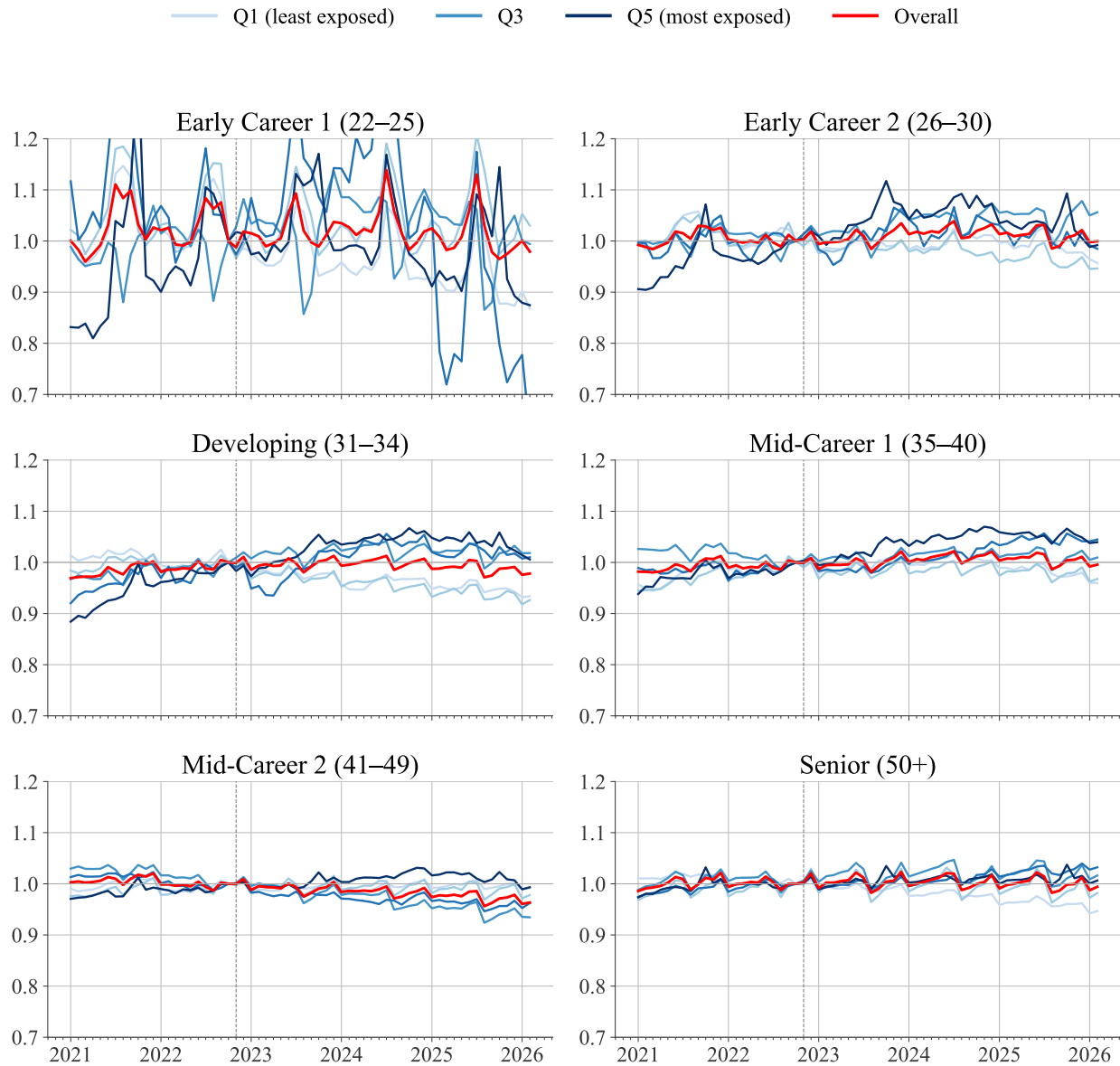
Employment per capita by age and AI-exposure quintile, Private sector (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1) in private sector, normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Exposure quintiles based on Eloundou et al. (2024) GPT-4 beta.

Figure A19: Employment by age and Eloundou et al. quintile, private sector. October 2022 = 1. The age-exposure gradient for 22–25 year-olds is visible in this sector.

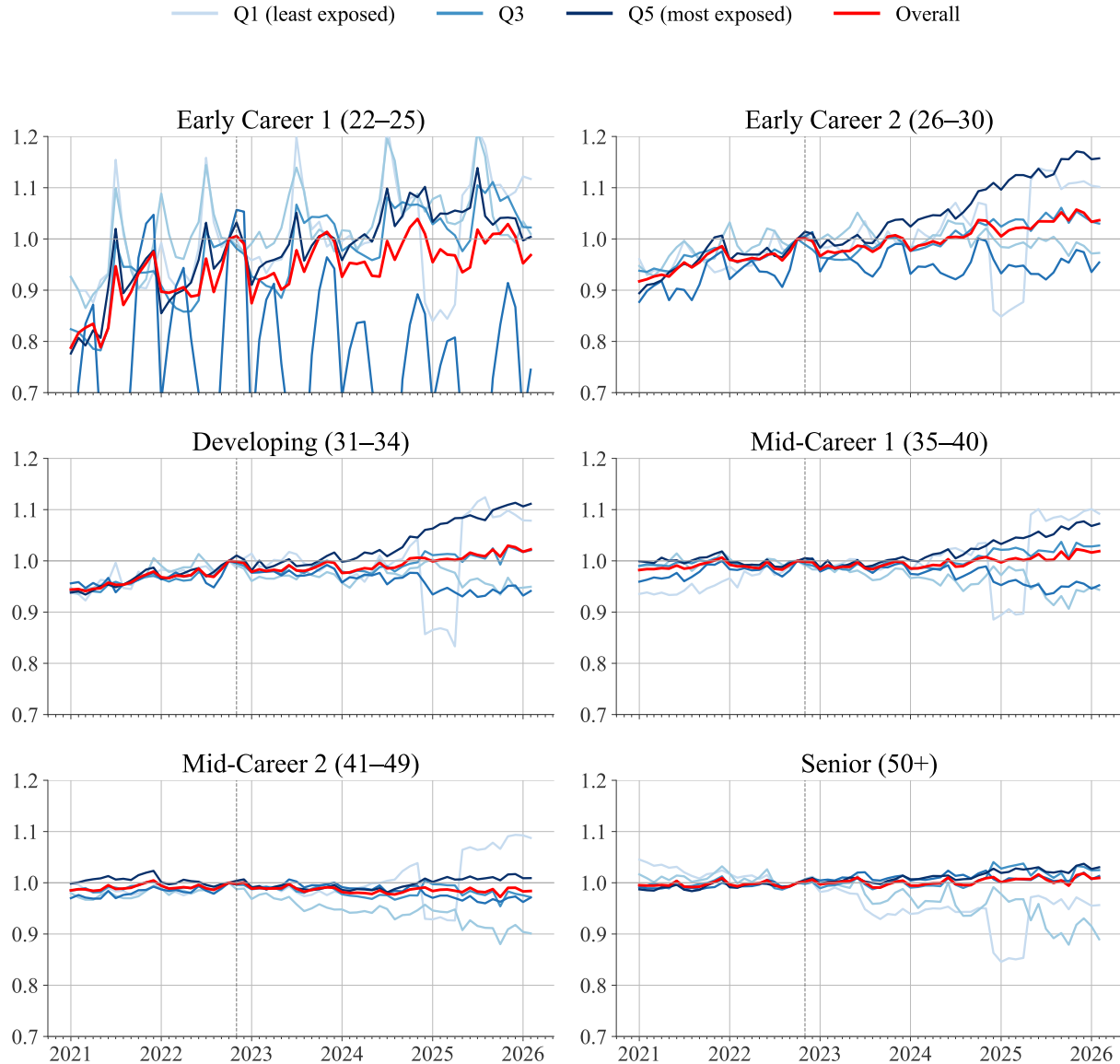
Employment per capita by age and AI-exposure quintile, Municipal sector (October 2022 = 1)



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1) in municipal sector, normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Exposure quintiles based on Eloundou et al. (2024) GPT-4 beta.

Figure A20: Employment by age and Eloundou et al. quintile, municipal sector. October 2022 = 1. No systematic quintile gradient in any age group.

Employment per capita by age and AI-exposure quintile, State sector (October 2022 = 1)



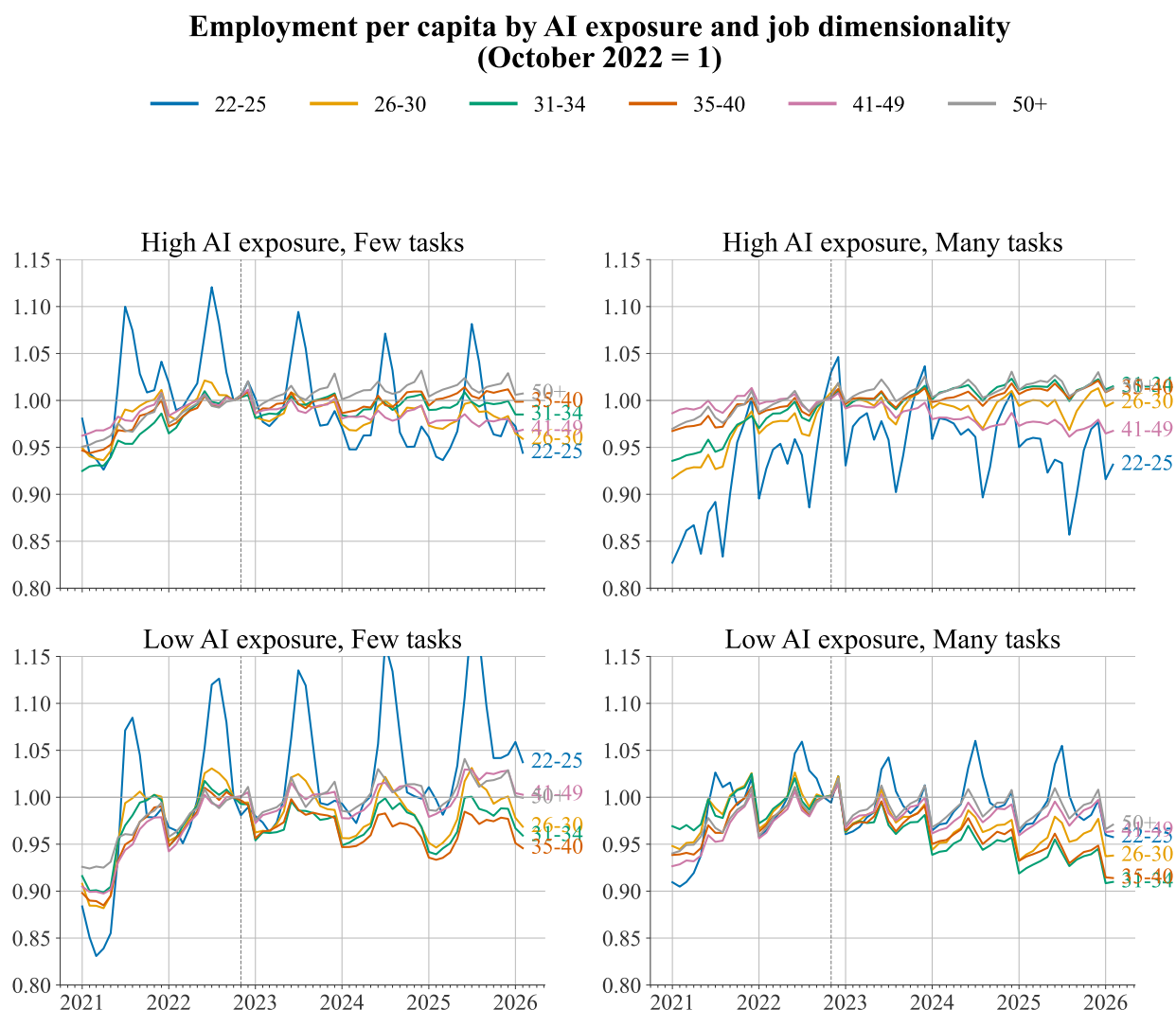
Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1) in state sector, normalized to October 2022 = 1. Dashed vertical line marks the launch of ChatGPT (November 2022). Exposure quintiles based on Eloundou et al. (2024) GPT-4 beta.

Figure A21: Employment by age and Eloundou et al. quintile, state sector. October 2022 = 1. No systematic quintile gradient in any age group.

al. exposure and on the number of O*NET task statements (median of 22 tasks), and plot normalized employment for workers aged 22–25 in each of the four quadrants. Residual SOC codes ending in “99” have artificially inflated task counts and are excluded from the task-count split.

Both high-exposure quadrants decline relative to the low-exposure quadrants, and the difference between high-exposure/few-tasks (−7.2%) and high-exposure/many-tasks (−8.4%) is modest (October 2022 to February 2026). The dominant margin is exposure level, not task count. Low-exposure occupations with few tasks grow (+1.9%).

The dimensionality split therefore provides limited descriptive support for the O-ring prediction in the current early-adoption window. We note the measurement caveats: a raw count of O*NET task statements does not capture the complementarity structure central to the O-ring model, the count may reflect coding granularity rather than true job complexity, and the median split is coarse. Alternative operationalizations (task-importance weighting, task similarity, or using the Handa et al. automation share as a proxy for substitutability) might deliver a different picture.



Notes: Employment per capita (SSB table 07459) with composition adjustment (ref: 2021-Q1), normalized to October 2022 = 1. Dashed line marks ChatGPT launch (November 2022). AI exposure: Eloundou et al. (2024) GPT-4 beta, median split. Dimensionality: O*NET task count per occupation, median split (22 tasks). Residual SOC categories excluded. O-ring prediction (Gans & Goldfarb 2025; Imas & Shukla 2026): decline concentrates in top-left panel.

Figure A22: Employment by AI exposure and job dimensionality, workers aged 22–25. Occupations are split at the median on Eloundou et al. exposure and on O*NET task count (22 tasks).